Revenue Optimal Bandwidth Allocation in a Class of Multihop Networks: Algorithms and Asymptotic Optimality

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***Abstract*— We study Bandwidth Reservation (BR) policies for the Bandwidth on Demand (BoD) problem in a class of multihop networks. We motivate an Erlang fixed-point BR heuristic for the general BoD problem by first establishing the optimality of BR on a class of multihop networks. The motivating problem is a wireline network comprising k links in tandem, each link of which is shared by two types of bandwidth demands, one type requiring one unit of bandwidth from every link, and the other type (being dedicated to the link) requiring one unit of bandwidth as well. First, for this k-hop tandem network, when each link has unit bandwidth, we demonstrate that a policy of BR form is optimal. We then study the BoD problem for a more general k-hop tandem network, in the "Kelly" limiting regime, where the arrival rates as well as the link bandwidths become large. For certain parameter regimes of the k-hop tandem network, we show that an admission control policy of BR form is asymptotically optimal. Motivated by these results, we propose an Erlang fixed-point based, link-by-link, heuristic algorithm for computing a BR policy for the BoD problem in a general network. We, finally, evaluate this proposal numerically.**

***Index Terms*— Software defined networking, bandwidth on demand, admission control, bandwidth reservation, asymptotic optimality, Erlang fixed-point approximation.**

I. Introduction

THE traditional Internet architecture combines control decisions (e.g., routing) in the same devices (e.g., routers) that buffer and forward the data packets, thus, effectively combining the "data plane" and the "control plane." In such an architecture, the distributed control has a myopic view of the network, often leading to suboptimal network performance [1]. Since 2008 (see, e.g., [2], [3], [4], [5]) SDN (software defined networking) has been studied, and experimented with, as an architecture that separates the control and data planes.[1] In this architecture, SDN controllers (computing devices connected to the network) have a global view of the network resources, and, therefore, can make better informed centralized decisions about resource allocation and network management as opposed to local routing decisions in the traditional architecture. Bandwidth on Demand (BoD) service is one of the many applications that can be provisioned using the SDN architecture, where the goal is to set up bandwidth guaranteed *virtual circuits* in a network [5]. The problem of "network slicing" in 5G networks is also one of allocating resources, on demand, albeit from a set of *nonhomogeneous* resources: link bandwidths, capacities of routers, and even compute resources. In this paper, motivated by the SDN context, we focus on the problem of dynamic allocation of multihop bandwidth to randomly arising requests in a general wireline network.[2]

We consider the following setup. Requests for bandwidth guaranteed *virtual circuits* arrive into the network over time. At an arrival epoch, based on the global network state, the SDN controller must decide whether to accept or reject that request. Once accepted, a virtual circuit is created in the network for a random "holding" time, and generates a certain revenue for the network operator. If a demand is rejected, the network operator does not gain any revenue, but bandwidth remains available for another, possibly higher revenue, demand that might soon arrive. As time progresses, depending on the decisions made by the controller at the arrival times of the requests (which we call a *policy*), revenue is generated. Thus, the SDN controller must implement an *admission control* policy that maximizes the long-run average revenue of the network.

Such a problem can be formulated as a Markov decision process (MDP); e.g., [7], [8], [9], [10]. However, finding an optimal policy is computationally hard for general networks due to the "very large" state space for the MDP[3] (this is referred to as the curse of dimensionality [7], [8, Section 4.2]). Although one can obtain the structure of an optimal policy for

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[1]Similar to Signaling System 7 in telephony [6] that separates *bearer channels* for voice and *data channels* for signaling.

[2]Bandwidth or bit-rate? We adopt a common misuse of terminology by using the term "bandwidth" when we actually mean "bit-rate".

[3]For instance, on a single link of 40 Gbps bandwidth with two types of requests, each having a bandwidth requirement of 10 Mbps, the cardinality of the state space is of the order of 10^6.

simple network topologies [9], [10], there are limited results available in the literature for general networks [11]. Hence, this motivates interest in the derivation and evaluation of revenue optimal admission control policies for the BoD problem in the context of SDNs, and the contributions of this paper are in this direction.

A. Our Contributions

We begin by recalling the MDP formulation of the BoD problem in a general SDN (Section II). The state space of this MDP is the set of all configurations of virtual circuits in the network, and the actions consist of state-dependent accept/reject decisions by the SDN controller upon arrival of requests. However, it is hard to solve this MDP directly due to the curse of dimensionality. Given this formulation of the BoD problem, there are three main contributions in this paper: We (i) solve the BoD problem for a k-hop tandem network[4] when each link has unit bandwidth, (ii) obtain an asymptotically optimal policy for this k-hop tandem network in the "Kelly" limiting regime under the condition that k-hop requests are of high priority and they are always accepted as long as there is sufficient bandwidth, and (iii) propose a link-by-link heuristic algorithm for the BoD problem in general networks and evaluate its performance numerically. We now highlight each of these contributions.

- We first consider a network that consists of k tandem links (see Figure 1), where each link is of unit bandwidth (we denote this by `Problem 0`). There are two types of requests: one type requiring unit bandwidth from every link, and another type requiring unit bandwidth from just the link to which it arrives. We solve the BoD problem in this network under a symmetry assumption (see Assumption (A0)) and demonstrate that a policy of *bandwidth reservation (BR)* form (see Definition 1) is optimal (Section III).
- For more general k-hop tandem networks (denoted by `Problem 1`), we then study the BoD problem in an asymptotic regime where the arrival rates of the requests and the link bandwidths become large (Section IV). This limiting regime is relevant in modern high-bandwidth networks [13], [14], [15]. It has also been studied in the context of loss networks [16], [17]. Under the condition that the k-hop requests are of high priority and, hence, are always accepted (see Assumption (A1)), we use a functional limit theorem [17] to establish the asymptotic optimality of a BR policy. This extends a similar result in the single link case [18] to a class of k-hop tandem networks. The key ingredient in the proof of this result is the unique identification of certain "acceptance probabilities" by analyzing suitable Markov chains associated with the vector of the residual bandwidths.
- Motivated by these results on the optimality of BR policies on a class of multihop networks, and the Erlang fixed-point approximation for computing blocking probabilities in loss networks (see [8, Chapter 5]), we then propose a BR policy for the BoD problem on general networks. This algorithm arrives at a policy by first computing certain optimal BR thresholds (via an iterative procedure that uses "thinned" arrival rates) on all the links separately, and then combining these to form a policy for the whole network[5] (Section V). A similar approach using a policy improvement procedure on each link was used in [20] and [21]. In our work, we further propose an optimization problem to obtain approximate BR thresholds on each link (Section V-B) for the k-hop tandem network under an additional symmetry assumption (see (A0)). Finally, we numerically evaluate the performance of the proposed BR-type policy (Section VI).

B. Review of the Literature

There has been a significant amount of published research on the problem of routing bandwidth guaranteed virtual circuits in a given network, even before the SDN architecture was proposed. In some studies, the objective was to maximize the probability of acceptance of a future arrival [22], [23], [24], whereas in others the objective was to maximize the minimum residual bandwidth [25], [26]. In this paper, our aim is to develop optimal BoD admission control to maximize the long-run average revenue for the network operator. We formulate this as an average cost MDP. Ross [8] used such an approach for the problem of admission control in loss networks. Although the structure of an optimal policy can be obtained for specific network topologies (e.g., see [10] for an access-backbone topology), due to the dimensional complexity [8, Section 4.2], there are very limited results available on the structure of an optimal admission control policy for general networks.

Bandwidth reservation type policies have been extensively studied in the context of admission control in loss networks. Lippman [9] showed the optimality of a trunk reservation type policy on a single link with multiple classes of calls having equal mean holding times and equal bandwidth requirements. It is also known that BR type policies need not be optimal on a single link when either the mean holding times are unequal or the bandwidth requirements are unequal [18, page 1060]. Bandwidth reservation policies have also been studied in the context of the Kelly limiting regime where the link bandwidths and the arrival rates scale linearly with a parameter N. In the case of a single link with two classes of requests with equal mean holding times and equal bandwidth requirements, Key [20, Section 5.2] showed that the optimal BR threshold scales as $\log N$. For general loss networks under bandwidth reservation controls, Hunt and Kurtz obtained a functional law of large numbers (FLLN) for the normalized call occupancy process [17]. Using the FLLN results in [17], Hunt and Laws [18] studied the problem of asymptotically optimal admission control on a single link with multiple classes of traffic (not necessarily having equal mean holding times or equal bandwidth requirements) and showed that a policy of trunk reservation form is asymptotically optimal. One of the

[4]This is a commonly used topology in SDNs, see, e.g., [12].

[5]The proposed algorithm does not re-route the requests, unlike in [19] and [11].

main contributions in this paper is to extend the results in [18] to the k-hop tandem network. We show that a BR policy with $\log N$ threshold is asymptotically optimal for the k-hop tandem network under certain parameter regimes (see (A1)). Motivated by this result, we then propose a heuristic BR policy for the BoD problem in general networks.

BR policies have also been studied in the context of the "critical loading" regime in loss networks. Reiman [27] considered a critically loaded link with two types of requests and showed that the optimal trunk reservation parameter scales as the square-root of the scaling parameter. In our work, we consider the k-hop tandem network under this critical loading and propose an optimization problem to identify the BR threshold on each link (Section V-B).

The problem of routing and admission control of bandwidth guaranteed virtual circuits was revived in the context of Software Defined Networks, a concept that became popular in the industry from around 2010 [3]. The SDN traffic engineering problem has been studied in [28] where the objective was to route the demands so as to minimize the delay and packet loss at the links. In [29], the authors proposed graph-based dynamic path computation algorithms to solve the traffic engineering problem in SDN. In the context of optical networks, [30], [31] modeled the BoD problem as an integer linear program and proposed heuristic algorithms to solve it. Similar problems have also been studied in the context of network slicing in 5G; see [32] for a two-level dynamic network slicing problem, and [33] for an end-to-end network slicing problem.

BoD formulations typically require the knowledge of the network topology and link bandwidths, and BoD traffic parameters such as the demand arrival rates. To avoid the need for such a priori knowledge, learning based online algorithms for the BoD problem in SDN have been developed. One of the earlier approaches in this direction was by Gokbayrak and Cassandras [34], where the goal was to minimize the weighted blocking probability of requests among a class of threshold-based policies, and the authors propose a stochastic approximation based algorithm to update the thresholds. In [35], the authors formulated the BoD problem as an integer program and evaluated an approach based on dynamically selecting one of several algorithms via an exploration-exploitation approach. In [36], the authors study a heuristic reinforcement learning based approach for the BoD problem with applications to 5G and IoT traffic. Their key idea was to use a pruning principle that resulted in an improved convergence of their heuristic learning procedure. For other related studies using reinforcement learning techniques, see [37], [38]. There have also been recent studies on demand prediction using deep learning techniques in the context of BoD, see [39], [40], [41], [42].

Unlike the above lines of research that propose learning algorithms to solve their MDP, in this paper, our focus is to obtain structural results from the asymptotic analysis of a BR policy for a specific tandem network model. Motivated by this result, we then propose a bandwidth reservation based heuristic policy for general networks. Our approach requires knowledge of the parameters (i.e., arrival rates and revenue rates), but these could be learned online, and the BR thresholds could be tuned online as well.

II. The Bandwidth on Demand Problem

In this section, we formulate the BoD problem in a general network as an MDP. Such formulations are well-known in the context of loss networks [7], [8]; we review this formulation here for completeness.

Let $\mathscr{G} = (V, E)$ be a network. Link j has C_j units of available bandwidth. There are several bandwidth request types, indexed by a finite set $\mathscr{R}$. Request type i is characterized by the five-tuple $(s_i, d_i, E_i, b_i, r_i)$ where s_i is the source node, d_i is the destination node, $E_i \subset E$ is a directed path from s_i to d_i through which request i will be routed, b_i is the bandwidth requirement from each link in the path, and r_i is the revenue rate when a virtual circuit of type i is active. Requests of type i arrive in a Poisson point process of rate λ_i and hold their required bandwidth for an exponentially distributed duration with rate μ_i; all arrival and departure processes are assumed to be independent of each other. A request of type i can be potentially accepted if the residual bandwidth on each link in E_i is at least b_i. If, depending on the admission control policy, a request for route i is accepted, a bandwidth of b_i units will be reserved on all links in E_i between the nodes s_i and d_i, and the network operator gains a reward of r_i per unit time until the virtual circuit leaves the network. In this setting, the network operator's objective is to maximize the long-run average revenue.

To quantify this objective, let $\mathbf{n}(t) = (n_1(t), n_2(t), \ldots, n_{|\mathscr{R}|}(t))$ denote the state of the network at a time t, where $n_i(t)$ is the number of virtual circuits of type i at time t. Note that $\mathbf{n}(t) \in \mathscr{N} = \{\mathbf{n} \in \mathbb{Z}_+^{|\mathscr{R}|} : \sum_{i \in \mathscr{R}} n_i(t) b_i 1_{\{j \in E_i\}} \leq C_j \text{ for all } j \in E\}$, where $\mathbb{Z}_+$ denotes the set of nonnegative integers. Let $\pi : \mathscr{N} \times \mathscr{R} \to \{0, 1\}$ be a given stationary deterministic policy, with the interpretation that, if a request arrives at time t when the network state is $\mathbf{n}$, $\pi(\mathbf{n})(i) = 1$ (resp. $\pi(\mathbf{n})(i) = 0$) implies that a request of type i is accepted (resp. rejected). Let $\{\mathbf{N}^{(\pi)}(t),\ t \geq 0\}$ be the controlled stochastic process describing the evolution of the system under a given policy π, where $N_i^{(\pi)}(t)$ denotes the number of virtual circuits of type i present at time t under the policy π. Define the long-run average revenue under a policy π by $g(\pi) := \limsup_{t \to \infty} \frac{1}{t} \mathsf{E}_\pi \left[\int_0^t \left(\sum_{i \in \mathscr{R}} N_i^{(\pi)}(u) r_i \right) du \right]$. In the BoD problem, the goal of the network operator is to find a policy π that attains $\sup_\pi g(\pi)$.

Since the state space $\mathscr{N}$ is finite, it is well-known from the theory of MDPs that there exists a stationary deterministic policy $\pi : \mathscr{N} \times \mathscr{R} \to \{0, 1\}$ that attains $\sup_\pi g(\pi)$ [43, Theorem 2, page 25]. However, obtaining the structure of an optimal policy is computationally hard [7], [8].

In this paper, we propose a policy of bandwidth reservation (BR) form for the BoD problem in general networks.

Definition 1: A policy is said to be of bandwidth reservation (BR) form[6] if it can be described by using a set of

[6]Also referred to as *trunk reservation* in circuit-switched networks.

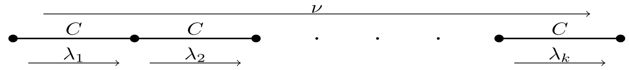

Fig. 1. A k-link line network with k one-hop requests and one k-hop request. C is the number of units of bandwidth on each link.

thresholds associated with each link and each request type that uses that link; an incoming request is accepted if the residual bandwidth on each link used by the request is at least the threshold associated with that request type on that link.

Example 1 (BR policy on a single link): Consider a single link of bandwidth C. There are two request classes, A and B, with thresholds t_A and t_B, respectively, and $0 < t_A < t_B < C$. When the residual bandwidth is greater than or equal to t_A (resp. t_B), a request of type A (resp. B) is accepted.

We motivate our BR policy in general networks by first showing the optimality (resp. asymptotic optimality) of a policy of BR form in a class of multihop networks when each link has unit bandwidth (resp. in a limiting regime where the link bandwidths and arrival rates become large). We next introduce this class of multihop networks.

A. The k-hop Tandem Network

In this paper, we study the BoD problem for a class of k-hop tandem networks which we now describe (see Figure 1). Most of the results of this paper are for this k-hop network (except the BR policy proposed in Section V, which is applicable to general networks).

There are k links in tandem. Link j has a bandwidth of C_j units. There are $k+1$ request types, a one-hop request on each link and a k-hop request that occupies all the links. Each type of request is assumed to have unit bandwidth requirement. The one-hop request on link i (we call this the type-i request) has arrival rate λ_i, and revenue rate r_i. The k-hop request has arrival rate ν and revenue rate r. We assume that the holding rates of all the requests are 1. In such k-hop tandem networks, under a further symmetry assumption, we first obtain the structure of an optimal policy when each link has unit bandwidth (Section III). This policy turns out to be of BR form. Obtaining the explicit structure of an optimal policy is not tractable when the link bandwidths are more than 1. Therefore, we consider an asymptotic regime where the link bandwidths and the arrival rates become large, and study asymptotically optimal policies in this limiting regime (Sections IV–V). Specifically, we show that a policy of BR form is asymptotically optimal.

III. Problem 0: Bandwidth Reservation Is Optimal for a k-Hop Tandem Network With Unit Bandwidth

In this section, we consider a simple situation where the BoD problem can be solved, and demonstrate that an optimal policy is of BR form. See Table I for the summary of the notation used in this section.

Recall the k-hop tandem network from Section II-A. We assume that that all links have unit bandwidth, i.e., $C_j = 1$ for all $1 \le j \le k$. Thus, any link can accommodate at most one virtual circuit at any time. We further assume the following symmetry assumption:

TABLE I

Summary of Notation Used in Section III

Symbol	Description
k	Number of links in the tandem network
λ	Arrival rate of one-hop requests
ν	Arrival rate of the k-hop requests
r	Revenue rate of the k-hop requests
$\mathcal{S}$	The state space; $\mathcal{S} = \{(i,j) : 0 \le i \le k, j \in \{0,1\}\}$
π	A policy; $\pi : \mathcal{S} \times \{1,2\} \to \{0(\text{Reject}), 1(\text{Accept})\}$ $\pi(s)(1)$ (resp. $\pi(s)(2)$) is the decision at state s for a one-hop (resp. k-hop) request
$\pi^{(0)}$	Always accept a one-hop request; always rejects a k-hop request
$\pi_l^{(1)}$	Always accept a k-hop request; reject a one-hop request at states $(i,0)$, $i \ge l$
$\pi_A^{(1)}$	Always accept a k-hop request; reject a one-hop request at states $(i,0)$, $i \in A$

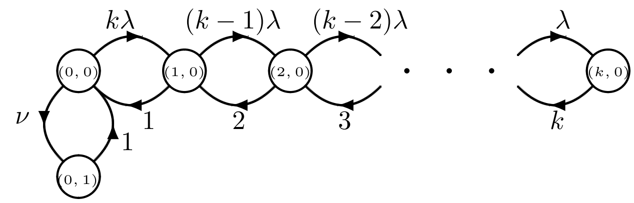

Fig. 2. The evolution of the process $X(t)$.

(A0) $\lambda_i = \lambda$ and $r_i = 1$ for all $1 \le i \le k$; $\mu_i = 1$ and $b_i = 1$ for all $1 \le i \le k+1$.

Let r denote the revenue rate of the k-hop request. Note that, under the assumption (A0), the revenue rate at a state $\mathbf{n}$ depends only on the number of ongoing one-hop virtual circuits in $\mathbf{n}$, and not the labels of the links occupied by them. Therefore, it suffices to consider policies that make admission control decisions based on the number of one-hop virtual circuits (this is also clear from the linear-programming approach to solve MDPs, see [8, page 132]). Hence, under the assumption (A0), the evolution of the state of the network can be equivalently described by the evolution of the two-tuple $X(t) = (i,j)$ where i (resp. j) is the number of one-hop (resp. k-hop) virtual circuits at time t, with appropriate transition rates. See Figure 2 for the evolution of the process $X(t)$ when no admission control is exercised. Using this equivalent description, we provide a characterization of an optimal policy. We first list the possible candidates for an optimal policy, and then prune the list further and compute the revenue under the remaining policies to find a best one.

Let $\mathcal{S}$ denote the state space of the process $X(t)$. Let $\pi : \mathcal{S} \times \{1,2\} \to \{0,1\}$ denote a policy; $\pi(s)(1) = 1$ (resp. $\pi(s)(2) = 1$) means that a one-hop request (resp. a k-hop request) is accepted at state s. Clearly, any policy π must satisfy the following constrains: $\pi(0,1)(1) = 0$, $\pi(0,1)(2) = 0$, $\pi(i,0)(2) = 0$ for $1 \le i \le k$, and $\pi(k,0)(1) = 0$. A k-hop request can be accepted only in $(0,0)$. Let $\pi^{(0)}$ denote the policy that always accepts a one-hop request and always rejects a k-hop request. Among the policies that accept a k-hop request at state $(0,0)$, there are many possible candidates for an optimal policy:

- $\pi_0^{(1)}$: $\pi_0^{(1)}(0,0)(2) = 1$ and always reject a one-hop request, i.e., $\pi_0^{(1)}(i,0)(1) = 0$ for $0 \le i \le k-1$.

- $\pi_l^{(1)}$, $1 \leq l \leq k$: $\pi_l^{(1)}(0,0)(2) = 1$, and reject a one-hop request if accepting that request will cause the number of virtual circuits to exceed l, i.e., $\pi_l^{(1)}(i,0)(1) = 1$ for $0 \leq i \leq l-1$, and $\pi_l^{(1)}(i,0)(1) = 0$ for $i \geq l$. Note that $\pi_k^{(1)}$ always accepts a request when there is sufficient bandwidth.
- The policy $\pi_A^{(1)}$ specified by a set $A \subseteq \{0, 1, \ldots, k\}$: $\pi_A^{(1)}$ rejects a one hop request whenever the number of one-hop virtual circuits is a member of A, i.e., $\pi_A^{(1)}(0,0)(2) = 1$, $\pi_A^{(1)}(i,0)(1) = 1$ whenever $i \notin A$, and $\pi_A^{(1)}(i,0)(1) = 0$ whenever $i \in A$. For example, $\pi_l^{(1)} = \pi_{\{l,2,\ldots,l\}}^{(1)}$.

Our main result in this section is the following theorem. Recall that r denotes the revenue rate of the k-hop request, λ (resp. ν) denotes the arrival rate of the 1-hop requests (resp. k-hop request).

Theorem 1: Assume (A0). Then an optimal policy for `Problem 0` *has the following structure:*

1) $\pi^{(0)}$ (i.e., do not accept k-hop requests, and always accept 1-hop requests if bandwidth is available) is optimal when $r \leq \frac{k\lambda}{\lambda+1}$,

2) $\pi_k^{(1)}$ (accept all requests if bandwidth is available) is optimal when $r \in \left[\frac{k\lambda}{\lambda+1}, \frac{k\lambda}{\lambda+1}\left(\frac{\nu+1}{\nu}\right)\left(\frac{(1+\lambda)^k}{(1+\lambda)^k-1}\right)\right]$,

3) $\pi_0^{(1)}$ (accept all k-hop requests when bandwidth is available, and no 1-hop requests) is optimal when $r \geq \frac{k\lambda}{\lambda+1}\left(\frac{\nu+1}{\nu}\right)\left(\frac{(1+\lambda)^k}{(1+\lambda)^k-1}\right)$.

Remark 1: Observe that when the revenue rate, r, from the k-hop request is small then it is optimal to accept only one-hop requests; on the other hand when the r exceeds a threshold, then it is optimal to accept *only* k-hop requests. Note, also, that we have a BR-type policy (see Figure 3 for its structure). The policies $\pi^{(0)}$, $\pi_k^{(1)}$, and $\pi_0^{(1)}$ correspond to the thresholds 1, 1, and 2 (resp. 2, 1, and 1) for the one-hop requests (resp. k-hop request) on all the links, respectively.

Proof: [Proof sketch] The proof involves two key ideas (see Appendix A for further details):

- *Pruning the set of candidate policies (Lemma 1):* Since the Markov chain $X(t)$ is a tree when no admission control is exercised (see Figure 2), among all policies that rejects a one-hop request at state $(i,0)$, we can restrict attention to policies that reject a one-hop request at all states $(j,0)$ for any $i \leq j \leq k$.
- *Monotonicity property (Lemma 3):* The average revenue of all the policies $\pi_l^{(1)}$, $1 \leq l \leq k-1$, lie below the average revenues of $\pi_0^{(1)}$ and $\pi_k^{(1)}$.

Using these steps, we conclude that an optimal policy must be one among $\{\pi^{(0)}, \pi_0^{(1)}, \pi_k^{(1)}\}$. We then compare the average revenues under these policies (see Lemma 2) to arrive at the structure of an optimal policy. □

Remark 2: We remark that the symmetry assumption (A0) is crucial in obtaining the structure of an optimal policy above. In particular, it helped us to (1) write down an equivalent description of the state of the k-hop network in terms of the number of one-hop and k-hop virtual circuits, and (2) show the monotonicity of the candidate policies (see Lemma 1), thereby yielding the structure of an optimal policy. Without this assumption it is not clear how to write down the structure of an optimal policy in terms of the parameters $\lambda_i, \mu_i, r_i, 1 \leq i \leq k$, and r, μ_{k+1}, and ν.

In summary, we demonstrated that a policy of BR form is optimal for the symmetric k-hop tandem network when each link has unit bandwidth. For general k-hop networks, solving the BoD problem becomes computationally hard, and it is difficult to obtain insights into an optimal policy. Further, one cannot expect to solve the general BoD problem (of Section II) unless some assumptions are imposed on the model. Therefore, it is natural to search for policies that are approximate solutions to the BoD problem. One way to do this is to study the BoD problem in an asymptotic regime where the link bandwidths and arrival rates become large, which is the topic of the next section.

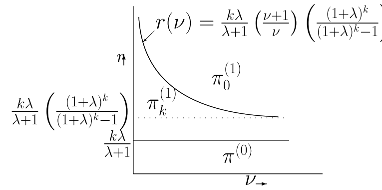

Fig. 3. Regions of optimality of various policies for `Problem 0` r–ν plane for fixed k and λ.

TABLE II

SUMMARY OF NOTATION USED IN SECTION IV

Symbol	Description
k	Number of links in the tandem network
C_j	The limit of the link bandwidth on link j scaled by N
λ_j	The limit of the arrival rate of one-hop requests on link j scaled by N
ν	The limit of the arrival rate of the k-hop requests scaled by N
r_j	Revenue rate of the one-hop requests on link j
r	Revenue rate of the k-hop requests
$\bar{\mathbf{x}}$	A solution to the LP in (3)
$\mathcal{R}^*$	The normalized optimal revenue defined in (2)
$\mathbf{x}^{(N)}(t)$	The scaled virtual circuit occupancy at time t
$m_j^{(N)}(t)$	The unscaled residual bandwidth process on j-th link in the N-th network
$P_j(\mathbf{x})$	The acceptance probability of a type-j request at state $\mathbf{x}$ in the large N limit of the scaled system
$\mathbf{m}_1^{(\mathbf{x})}, \mathbf{m}_2^{(\mathbf{x})}$	The Markov processes used to define the acceptance probabilities

IV. `Problem 1`: Asymptotic Optimality of Bandwidth Reservation in k-hop Tandem Networks

In this section, we study asymptotically optimal policies for the k-hop tandem network in the Kelly limiting regime [44] where the link bandwidths and arrival rates become large. Specifically, we show that a bandwidth reservation type policy is asymptotically optimal for the k-hop tandem network under certain parameter regimes.

A. The Setting and the Objective of `Problem 1`

We consider a sequence of k-hop tandem networks indexed by $N \geq 1$. For the N-th network, let $C_i(N)$ denote the link bandwidth of the ith link in the network, and $\lambda_i(N)$ (resp. $\nu(N)$) denote the arrival rates of type-i requests (resp. k-hop requests). We assume that all the holding times are $\exp(1)$ random variables (i.e., $\mu_i(N) = 1$ for all i and N), and that all requests have unit bandwidth requirements. The revenue rate of the one-hop requests on link j is r_j; see Table II for a summary of the notation. We consider the following asymptotic regime:

$$\frac{C_i(N)}{N} \to C_i,\ \frac{\lambda_i(N)}{N} \to \lambda_i, 1 \le i \le k, \text{ and } \frac{\nu(N)}{N} \to \nu, \quad (1)$$

as $N \to \infty$. Such asymptotic regimes are relevant in the context of SDNs with large link bandwidths [13], [14], [15], and were introduced in the context of the classical circuit switched loss networks [16], [17]. For the N-th network, let $\mathcal{R}^{(N)}(\pi)$ denote the average revenue under a stationary deterministic policy π. Under the above scaling, it is easy to see that the optimal revenue scales as $O(N)$. Define the normalized optimal revenue

$$\mathcal{R}^* := \liminf_{N\to\infty} \sup_{\pi\in\Pi^{(N)}} \frac{1}{N}\mathcal{R}^{(N)}(\pi), \quad (2)$$

where $\Pi^{(N)}$ is the set of all stationary deterministic policies for the N-th network. In this section, our goal is to propose a policy for the N-th network and show that, in certain parameter regimes, its normalized revenue is close to $\mathcal{R}^*$ for large enough N. We introduce the following definition.

Definition 2: A sequence of policies $\{\pi^{(N)}\}$ is said to be asymptotically optimal if $\liminf_{N\to\infty} \frac{1}{N}\mathcal{R}^{(N)}(\pi^{(N)}) = \mathcal{R}^*$.

The goal of this section is to show that, under certain parameter regimes, a sequence of policies of BR form is asymptotically optimal for the k-hop tandem network under the parameter scaling in (1). Towards this, we proceed via the following steps; see [18]:

1) Find an upper bound for $\mathcal{R}^*$ using a linear-program (LP);
2) For each N, describe a BR policy using the solution of this LP;
3) Characterize the limiting dynamics of the k-hop network under the sequence of proposed policies in the asymptotic regime;
4) Show that the above dynamics converges to a fixed point;
5) Show that the above fixed point is a solution to the LP, and, hence, conclude that the BR policy proposed in Step 2 is asymptotically optimal.

The main novelty of our result is the unique identification of the acceptance probabilities in Step 3, and thereby uniquely characterizing the limiting dynamical evolution of the network. This identification is performed via studying certain multidimensional Markov chains. In the rest of this section, we proceed via the above five steps to show that a policy of BR form is asymptotically optimal for the k-hop tandem network under certain parameter regimes.

B. An LP-Based Upper Bound

We first derive an upper bound for $\mathcal{R}^*$ in (2) using an LP. Consider the N-th network under a policy π. Since the finite-state Markov chain $\{\mathbf{n}(t), t \ge 0\}$ under π that describes the evolution of the number of ongoing virtual circuits is irreducible, it possesses a unique stationary distribution. Under stationarity, let $x_j^{(N)}, 1 \le j \le k$, denote the mean number of one-hop virtual circuits on link j and $x_{k+1}^{(N)}$ denote the mean number of k-hop virtual circuits. Then the time-average revenue under π equals $\sum_{j=1}^k r_j x_j^{(N)} + r x_{k+1}^{(N)}$. Since the number of virtual circuits on a link cannot exceed the link bandwidth, we have $x_j^{(N)} + x_{k+1}^{(N)} \le C_j(N), 1 \le j \le k$. By Little's theorem (e.g., [45, page 276]), we have $0 \le x_j^{(N)} \le \lambda_j(N), 1 \le j \le k$, and $0 \le x_{k+1}^{(N)} \le \nu(N)$. Normalizing both the revenue and the constraints by N, letting $N \to \infty$ and using (1), $\mathcal{R}^*$ is upper bounded by the optimum value of the following LP (see [18] for details).

$$\text{maximize} \sum_{j=1}^k r_j x_j + r x_{k+1}, \quad (3)$$

$$\text{subject to } x_j + x_{k+1} \le C_j, \quad j = 1, 2, \ldots, k, \quad (4)$$

$$0 \le x_j \le \lambda_j, \quad j = 1, 2, \ldots, k, \quad (5)$$

$$0 \le x_{k+1} \le \nu. \quad (6)$$

Note that this LP has $k+1$ unknowns and $3k+2$ inequality constraints, and it does not depend on the parameter N. It can be solved directly by proposing a solution and checking the complementary slackness conditions, or by using the simplex method. We make the following assumption on the solution of the above LP.

(A1) For any solution $\bar{\mathbf{x}}$ of the LP (3), we have $\bar{x}_{k+1} = \nu$.

This assumption has the interpretation that k-hop requests have sufficiently large revenue rates compared to the one hop requests. However, note that this assumption is indeed true for many choices of the parameters of the problem, irrespective of the revenue rates. For instance, when $\lambda_j + \nu \le C_j$ for all j, then (A1) holds irrespective of the revenue rates. Even when such a condition is not true, it is reasonable to assume that a virtual circuit requesting more network resources would generate more revenue. Indeed, analogous assumptions were used in the context of network slicing in 5G: for instance, [46] assumed that "inelastic traffic" generate more revenue than "elastic traffic" because of stringent requirements for network resources, and the results of a pricing problem in [47] suggested that users pay more in macro-cells compared to small-cells. As a result of this assumption, the BR policy that we propose always accepts a k-hop request whenever there is sufficient bandwidth on all the links. This assumption makes tractable the analysis of certain multidimensional Markov chains used in the identification of the "acceptance probabilities" in (9), and thereby leads to the unique identification of the limiting dynamics of the k-hop network. For further discussion on this assumption, see Section IV-F.

Under (A1), we can write down the solutions of the LP (3).

Proposition 1: Let $\bar{\mathbf{x}}$ denote a solution to the LP (3) and assume (A1). Then, $\bar{\mathbf{x}}(j) = \min\{C_j - \nu, \lambda_j\}$, $j \in \{1, 2, \ldots, k\}$, and $\bar{x}_{k+1} = \nu$.

Remark 3: In the context of restless dynamic multi-class bandits, a similar set of steps as in Section IV-A were followed in [48] to show the asymptotic optimality of Whittle's index policy under a limiting regime where the number of active bandits and the arrival rates of bandits become large. There, the limiting dynamics was assumed to have a global attractor (see Condition 4.10 in [48]) whereas we show this for our problem in Step 4 of the procedure outline in Section IV-A. Also, the analog of Step 3 (Section IV-A) was carried out in [48] by assuming certain properties of the stationary distribution of the prelimit (see Condition 4.12 in [48]), whereas we use assumption (A1) on solution(s) of the LP in (3).

C. A Bandwidth Reservation Policy

Given the parameters of the problem, the LP in (3) can be solved, but this does not tell us how to perform a time average revenue maximizing admission control as the different types of requests arrive. We now propose an admission control policy of the bandwidth reservation (BR) type, for the N-th network.

Fix a solution $\bar{\mathbf{x}}$ of the LP in (3). Let $n_j^{(N)}(t)$, $1 \leq j \leq k+1$, denote the number of ongoing virtual circuits of type-j requests in the N-th network. Consider the residual bandwidth process on the j-th link in the N-th network:

$$m_j^{(N)}(t) := C_j(N) - (n_j^{(N)}(t) + n_{k+1}^{(N)}(t)). \tag{7}$$

The bandwidth reservation policy:

At time t,

- A k-hop request is accepted when $m_j^{(N)}(t) \geq 1$ for each $1 \leq j \leq k$;
- A one-hop request on link j is accepted when $m_j^{(N)}(t) \geq \log N$.

This policy has the following interpretation. Assumption (A1) implies that the revenue rate of k-hop requests is large enough that we accept it as long as there is enough bandwidth in the network to accommodate that request. On the other hand, a type-j request is accepted whenever the free bandwidth on the j-th link is at least $\log N$ (the threshold $\log N$ for the one-hop requests can be replaced by any function $r(N)$, as long as $r(N) \to \infty$ as $N \to \infty$ and $r(N)$ is $o(N)$; see also [18]). The fact that the BR threshold is $o(N)$ and that it goes to ∞ as $N \to \infty$ enables us to prove the asymptotic optimality of the proposed policy, i.e., the average revenue rates given by the solution of the LP in (3) (which are actually upper bounds) are achieved as $N \to \infty$. Further, under (A1), this policy does not require the knowledge of the arrival rates or revenue rates of the requests.

D. The Limiting Dynamics

Consider the N-th network and define the scaled virtual circuit occupancy processes

$$x_j^{(N)}(t) := \frac{n_j^{(N)}(t)}{N}, \ j \in \{1, 2, \ldots, k+1\}. \tag{8}$$

$\{\mathbf{x}^{(N)}\} = \{\mathbf{x}^{(N)}(t), t \geq 0\}$ is a Markov process on $\mathcal{X} = \{(x_1, \ldots, x_{k+1}) : x_i + x_{k+1} \leq C_i \ \forall i\}$. We shall view $\{\mathbf{x}^{(N)}\}$ as a random element of the space of trajectories $D_{\mathcal{X}}[0, \infty)$.[7] We now study a law of large numbers for $\{\mathbf{x}^{(N)}\}$.

We first introduce some notation. Let $B_j = \{\mathbf{x} \in \mathcal{X} : x_j + x_{k+1} = C_j\}, 1 \leq j \leq k$, denote a boundary of the state space $\mathcal{X}$; this is the set where the j-th link can possibly reject a request in the limiting system. Let $\mathbf{e}_j$, $1 \leq j \leq k$, denote the k-length vector with a 1 at the j-th position and a 0 everywhere else, and let $\mathbf{1}$ denote the all-1 vector of length k.

We now state the convergence theorem for $\{\mathbf{x}^{(N)}\}$.

[7] $D_{\mathcal{X}}[0, \infty)$ is the space of $\mathcal{X}$-valued functions on $[0, \infty)$ that are right continuous with left limits, equipped with the Skorohod metric [49, Chapter 3].

Theorem 2: For `Problem 1`, suppose that $\mathbf{x}^{(N)}(0)$ converges weakly to $\mathbf{x}(0)$ as $N \to \infty$. Then, under Assumption (A1), $\{\mathbf{x}^{(N)}\}$ converges weakly, in $D_{\mathcal{X}}[0, \infty)$, to a continuous trajectory $\mathbf{x}(t) = (x_j(t), 1 \leq j \leq k+1)$, $t \geq 0$, that satisfies

$$x_j(t) = x_j(0) + \int_0^t (\lambda_j P_j(\mathbf{x}(u)) - x_j(u)) du, \tag{9}$$

$j \in \{1, 2, \ldots, k+1\}$ and $t \geq 0$, where $\lambda_{k+1} = \nu$, and the acceptance probabilities $P_j(\mathbf{x}), j \in \{1, 2, \ldots, k+1\}$ and $\mathbf{x} \in \mathcal{X}$, are defined as follows:

- $P_{k+1}(\mathbf{x}) := 1$ for all $\mathbf{x} \in \mathcal{X}$;
- For each $j \in \{1, 2, \ldots, k\}$, if $x_j + x_{k+1} < C_j$ then $P_j(\mathbf{x}) = 1$. If $x_j + x_{k+1} = C_j$ then

$$P_j(\mathbf{x}) := \left(\frac{C_j - \nu}{\lambda_j}\right) \wedge 1. \tag{10}$$

Remark 4: This result has the following interpretation. Since the revenue from k-hop requests are sufficiently large (by (A1)), the limiting dynamical system (9) always accepts k-hop requests (i.e. $P_{k+1}(\mathbf{x}) = 1$ for all $\mathbf{x} \in \mathcal{X}$). For one-hop requests, when there is sufficient bandwidth on link j (which, under the large N scaling, just corresponds to $x_j + x_{k+1} < C_j$), all type-j requests are accepted. At the boundary, the acceptance probability is determined based on the residual bandwidth on link j. If the residual bandwidth is large compared to the arrival of type-j requests (i.e., when $\lambda_j \leq C_j - \nu$), then all type-j requests are accepted (which yields $P_j(\mathbf{x}) = 1$). Otherwise, (i.e., when $\lambda_j > C_j - \nu$), type-j requests are accepted such that the link bandwidth is not exceeded (which results in $P_j(\mathbf{x}) = (C_j - \nu)/\lambda_j$). The form of $P_j(\mathbf{x})$ in the above theorem is obtained as a consequence of the threshold $\log N$ in our proposed policy. The fact that this threshold is $o(N)$ results in $P_j(\mathbf{x}) = 1$ away from the boundary B_j, and the fact that $\log N \to \infty$ as $N \to \infty$ results in the form of $P_j(\mathbf{x})$ on the boundary B_j.

Remark 5: The scaling and limiting regime considered in this paper is different from the Iglehart-Borovkov limit (e.g., [50]). The latter considers a central limit theorem type scaling which yields a diffusion limit (thus, retaining stochastic fluctuations), whereas the scaling and limiting regime considered in this paper yields a fluid limit, and is motivated by the seminal work of Hunt and Kurtz [17].

The interplay between the $\mathbf{m}^{(N)}$ and $\mathbf{x}^{(N)}$ processes: Note that, in a given time duration, the process $\mathbf{m}^{(N)}$ makes $O(N)$-many constant-size jumps, whereas, owing to the scaling by N, the process $\mathbf{x}^{(N)}$ makes $O(N)$-many $O(1/N)$-size jumps. Therefore, in a small time duration, $\mathbf{x}^{(N)}$ changes by $O(1)$ whereas $\mathbf{m}^{(N)}$ changes by $O(N)$. To obtain the acceptance probabilities in the fluid limit in (9), we must study the process $\mathbf{m}^{(N)}$ when $\mathbf{x}^{(N)}(t)$ is frozen at some state $\mathbf{x} \in \mathcal{X}$. We now discuss the behavior of $\mathbf{m}^{(N)}$ for large N which plays a crucial role in the proof of Theorem 2 (see also [17], [18]).

We first consider the k-hop request. Consider the N-th system. If we freeze $\mathbf{x}^{(N)}(t)$ to some $\mathbf{x} \in \mathcal{X}$, then the residual bandwidth process $\mathbf{m}^{(N)}(t)$ acts as a Markov process on $\mathbb{Z}_+^k$. Its transition rates are as follows: a departure of a type-j (resp. k-hop) virtual circuit occurs at rate Nx_j (resp. Nx_{k+1}) that results in the transition $\mathbf{m} \to \mathbf{m} + e_j$ (resp. $\mathbf{m} \to \mathbf{m} + \mathbf{1}$), and when $\mathbf{m}_j^{(N)}(t) \geq \log N$ (resp. $\mathbf{m}_i^{(N)}(t) \geq 1$ for all i), an arrival of a type-j (resp. k-hop) request occurs at rate $\lambda_j(N)1_{\{\mathbf{m}_j \geq \log N\}}$ (resp. ν) that results in the transition $\mathbf{m} \to \mathbf{m} - e_j$ (resp. $\mathbf{m} \to \mathbf{m} - \mathbf{1}$). Note that all these transition rates are $O(N)$; therefore, for large N, over very short periods of time, the free bandwidth process $\mathbf{m}^{(N)}$ equilibrates quickly. We therefore anticipate that the stationary behavior of $\mathbf{m}^{(N)}$ will play a role in defining the vector-field of the limiting behavior of the slow process $\mathbf{x}^{(N)}$. To study its stationary behavior, noting that the threshold $\log N \to \infty$ as $N \to \infty$, we define a time-scaled (i.e., reducing all the transition rates by a factor of N) process $\mathbf{m}_1^{\mathbf{x}}$ on $(\mathbb{Z}_+ \cup \{+\infty\})^k$ using the following transition rates.

$$\mathbf{m}_1^{\mathbf{x}} \to \begin{cases} \mathbf{m}_1^{\mathbf{x}} + \mathbf{1} & \text{at rate } x_{k+1}, \\ \mathbf{m}_1^{\mathbf{x}} + \mathbf{e}_j & \text{at rate } x_j, \ \text{ for all } j \in \{1, 2, \ldots, k\}, \\ \mathbf{m}_1^{\mathbf{x}} - \mathbf{1} & \text{at rate } \nu \times 1_{\{\mathbf{m}_1^{\mathbf{x}}(j) \geq 1 \forall j \in \{1,2,\ldots,k\}\}}. \end{cases} \tag{11}$$

Since the transition rates of the process $\mathbf{m}_1^{\mathbf{x}}$ defined above are scaled versions of that of $\mathbf{m}^{(N)}$ (which corresponds to time scaling of $\mathbf{m}^{(N)}$ by a factor of $1/N$), the stationary behavior of $\mathbf{m}^{(N)}$ can be described using that of the process $\mathbf{m}_1^{\mathbf{x}}$ above. Studying the stationary behavior of $\mathbf{m}_1^{\mathbf{x}}$ enables us to show that $P_{k+1}(\mathbf{x}) = 1$ for all $\mathbf{x} \in \mathcal{X}$.

Now consider the one-hop request on link j. Since our BR threshold is $\log N$ which goes to ∞ as $N \to \infty$, the acceptance of a type-j request is not reflected in the process $\mathbf{m}_1^{\mathbf{x}}$ defined above, which is a time-scaled version of the process $\mathbf{m}_1^{(N)}$. To capture the reservation based admission control of type-j requests, we consider the process $\mathbf{m}^{(N)} - (\mathbf{1} \times \lfloor \log N \rfloor)$ in addition to $\mathbf{m}^{(N)}$. Towards this, define the process $\mathbf{m}_2^{\mathbf{x}}$ on $(\mathbb{Z} \cup \{+\infty, -\infty\})^k$ with the following transition rates:

$$\mathbf{m}_2^{\mathbf{x}} \to \begin{cases} \mathbf{m}_2^{\mathbf{x}} + \mathbf{1} & \text{at rate } x_{k+1}, \\ \mathbf{m}_2^{\mathbf{x}} + \mathbf{e}_j & \text{at rate } x_j, \ \text{ for all } \ j \in \{1, 2, \ldots, k\}, \\ \mathbf{m}_2^{\mathbf{x}} - \mathbf{1} & \text{at rate } \nu, \\ \mathbf{m}_2^{\mathbf{x}} - \mathbf{e}_j & \text{at rate } \lambda_j \times 1_{\{\mathbf{m}_2^{\mathbf{x}}(j) \geq 0\}}, \\ & \qquad \text{for all } j \in \{1, 2, \ldots, k\}. \end{cases} \tag{12}$$

This is a time-scaled version of $\mathbf{m}^{(N)} - (\mathbf{1} \times \lfloor \log N \rfloor)$ (by a factor $1/N$). A type-j request is accepted when $m_j^{(N)} - \log N \geq 0$, which can be captured using the process $\mathbf{m}_2^{\mathbf{x}}$ defined above.

Note that the processes $\mathbf{m}_1^{\mathbf{x}}$ and $\mathbf{m}_2^{\mathbf{x}}$ are defined on the spaces $(\mathbb{Z}_+ \cup \{+\infty\})^k$ and $(\mathbb{Z} \cup \{+\infty, -\infty\})^k$, respectively. The compactification of the state spaces are essential to describe the limiting behavior of the process $\{\mathbf{x}^{(N)}\}$, as we shall see. This also results in multiple closed communicating classes for the processes $\mathbf{m}_1^{\mathbf{x}}$ and $\mathbf{m}_2^{\mathbf{x}}$. For instance, if one of the components of $\mathbf{m}_1^{\mathbf{x}}$ is initialized at $+\infty$, then it stays at $+\infty$ forever. As a result, these processes could possess multiple stationary distributions. This is the main difficulty in the unique identification of the acceptance probabilities in (9).

See the appendix for the proof of Theorem 2. The main novelty in the proof is the unique identification of the acceptance probabilities $P_j(\mathbf{x})$, $1 \leq j \leq k+1$ and $\mathbf{x} \in \mathcal{X}$. To identify $P_{k+1}(\mathbf{x})$ at a boundary state $\mathbf{x}$, we consider multidimensional Markov chains $\mathbf{m}_1^{\mathbf{x}}$ and $\mathbf{m}_2^{\mathbf{x}}$ defined in (11) and (12) respectively, and argue that they are either null recurrent or transient. To identify $P_j(\mathbf{x})$, $1 \leq j \leq k$, at a boundary state $\mathbf{x}$, we argue that the limiting dynamics must always belong to the set $\mathcal{X}$. The unique identification of these acceptance probabilities, together with the results in [17], enables us to prove the convergence of $\{\mathbf{x}^{(N)}\}$.

E. Convergence of the Limiting Dynamics

Recall $\bar{\mathbf{x}}$, a solution of the LP (3).

Proposition 2: Let $\{\mathbf{x}(t), t \geq 0\}$ be a trajectory satisfying (9). Then $\mathbf{x}(t) \to \bar{\mathbf{x}}$ as $t \to \infty$, where $\bar{\mathbf{x}}$ is a solution to the LP in (3).

See Appendix B for a proof. By the above result, all trajectories of the limiting dynamics (9) converge to $\bar{\mathbf{x}}$, the solution of the LP (3) that we started with. We have thus argued Steps 1–5 in Section IV-A under the assumption (A1). Therefore we conclude that the policy proposed in Section IV-C, which is of BR form, is asymptotically optimal for the k-hop tandem network under (A1).

F. Discussion

We now highlight some difficulties when (A1) is not enforced. Consider a 2-hop network. Let $\bar{\mathbf{x}}$ be a solution to the LP (3) and suppose that $0 < \bar{x}_3 < \nu$, $\bar{x}_j = \lambda_j$, $j = 1, 2$, and $C_1 = C_2 = C$. In this case, for the N-th network, it is natural to consider the BR policy that accepts a 2-hop request when the residual bandwidths on both the links are at least $\log N$. The functional law of large numbers results in [17] are still applicable; we will now get subsequential convergence of the process $\{\mathbf{x}^{(N)}\}$ to a limiting dynamics whose driving vector-field involves the acceptance probabilities. *The difficulty lies in the unique identification of these acceptance probabilities.* Consider a state $\mathbf{x}$ such that $x_i + x_3 = C$, $i = 1, 2$. To identify the acceptance probability of a 2-hop request in $\mathbf{x}$, we are faced with the problem of identifying the stationary distributions of the Markov chain $\mathbf{m}^{\mathbf{x}}$ on $(\mathbb{Z} \cup \{+\infty, -\infty\})^2$ with the following transition rates:

$$\mathbf{m}^{\mathbf{x}} \to \begin{cases} \mathbf{m}^{\mathbf{x}} + \mathbf{1} & \text{at rate } x_3, \\ \mathbf{m}^{\mathbf{x}} + \mathbf{e}_j & \text{at rate } x_j, \ j \in \{1, 2\}, \\ \mathbf{m}^{\mathbf{x}} - \mathbf{1} & \text{at rate } \nu \times 1_{\{\mathbf{m}^{\mathbf{x}}(1) \geq 0, \, \mathbf{m}^{\mathbf{x}}(2) \geq 0\}}, \\ \mathbf{m}^{\mathbf{x}} - \mathbf{e}_j & \text{at rate } \lambda_j, \ j \in \{1, 2\}. \end{cases}$$

It is a priori not clear if the above chain restricted to $\mathbb{Z}^2$ is positive recurrent, null recurrent, or transient. Clearly, owing to the compactification of the state space, the chain possesses stationary distributions; for instance, the point mass at $(+\infty, +\infty)$ is a stationary distribution of the above chain. If the chain restricted to $\mathbb{Z}^2$ turns out to be positive recurrent, it will possess a stationary distribution supported on $\mathbb{Z}^2$. Therefore, it is not clear which stationary distribution of this chain plays a role in defining the acceptance probability of a 2-hop request in the state $\mathbf{x}$. The situation becomes more difficult in higher dimensions, and for problems that do not possess any symmetry. Because of this difficulty in the identification of the acceptance probabilities, one cannot, in general, conclude the convergence of the sequence $\{\mathbf{x}^{(N)}\}$. In the proof of Theorem 2, assumption (A1) enabled us to show that the Markov chain $\mathbf{m}_1^{\mathbf{x}}$ in (11) restricted to $\mathbb{Z}^2$ cannot possess a stationary distribution, which led to the unique identification of the acceptance probability of a 2-hop request.

Some refinements of the Hunt and Kurtz theory towards the identification of the acceptance probabilities have been done in [51] and [52]. However it is not clear how one can apply them to our case. While it is natural to conjecture that a suitable BR policy is asymptotically optimal for the k-hop problem in all the parameter regimes, a rigorous proof is still lacking.

Remark 6: Non-asymptotic analysis: It would be interesting to study the k-hop network in the non-asymptotic regime, i.e., to propose a threshold-based policy for a fixed network, and possibly get a bound on the difference between the optimal revenue and the average revenue under the proposed policy in terms of the network parameters. This non-asymptotic analysis is beyond the scope of this paper, and it is an interesting future direction.

In summary, we have proved that a policy of BR form is asymptotically optimal for the k-hop tandem network in the limiting regime where the link bandwidths and the arrival rates become large, under the additional assumption that k-hop requests are of high priority and they are accepted whenever there is sufficient bandwidth on all the links. To extend this result to general networks (and to the k-hop tandem network when the k-hop requests are not of high priority), the key difficulty is in the unique identification of the acceptance probabilities that appear in the limiting dynamics. The asymptotic optimality in this section suggests that one can employ the proposed BR policy and obtain a revenue that becomes "close" to the optimal revenue in the "Kelly" limiting regime. Motivated by this result for the k-hop tandem network, we next propose a BR policy for the BoD problem in a general network.

V. General Networks: An Erlang Fixed-Point Heuristic for a Bandwidth Reservation Policy

The results of [9], [18], and [19], and our results in the previous sections motivate us propose a policy of BR form for the BoD problem in general networks (i.e., for the BoD problem formulated in Section II). In this section, we first propose a BR policy to solve the BoD problem in a general network using a link-by-link heuristic algorithm. This policy involves the computation of certain BR thresholds on each link in the network, and then combining these thresholds on each link to form a policy for the network. We then study certain approximations of these thresholds for large values of arrival rates and link bandwidths.

We remark that the proposed policy does not re-route the requests, unlike in [19] and [11]. The routes of all the requests are predetermined, and the goal of the SDN controller is to implement an admission control to maximize the long-run average revenue.

A. `BR-EFP`: a Policy for General Networks (without Assuming (A0) or (A1))

Recall the BoD problem from Section II. The proposed `BR-EFP` policy is based on the Erlang Fixed-Point (EFP) approximation. We assume that (i) the links block requests independently, and (ii) the arrival process to a link, thinned by the blocking probabilities on the other links, is Poisson. Such approximations have been considered in loss networks, see [8], [44].

Based on these two assumptions, we now propose our iterative scheme to arrive at a policy for a general network. Since every request requires equal amount of bandwidth from each link in its route, we assume that the revenue rate of each request is equally divided among the links that it uses. Then, in a link-by-link approach, if we account for the revenue of each admitted request from each link along its path, we correctly account for the revenue from that request. We consider link j in isolation. Motivated by the optimality of a BR policy on a single link under equal mean holding times and equal bandwidth requirements[8] [9], in the first iteration, we identify an optimal BR threshold for each request type i that uses link j (i.e., we find an optimal policy among the class of BR policies[9]). We repeat this for all links j. Using these reservation thresholds, we compute the blocking probabilities of each request type i on all links $j \in E_i$. Using these, by assuming that the links block requests independently, we compute the "thinned" arrival rate of request type i to link j by multiplying the actual arrival rate of request i with the product of the acceptance probabilities of request i on all the links other than link j. We repeat this for all request types i and links j. This yields a new set of arrival rates for each request type i on each link $j \in E_i$. We then compute a new set of reservation thresholds on each link based on these new arrival rates, and repeat this procedure until convergence. The link-by-link heuristic algorithm gives us a BR threshold on each link in the network. Once we have this, we build the `BR-EFP` policy for the network as follows.

> *A type i request is accepted at state $\mathbf{n}$ if, for all links $j \in E_i$, the BR threshold on link j for request i accepts this request on link j.*

For a `Python` implementation of this policy, see [53].

Remark 7: Unknown parameters: If the arrival and departure rates of the bandwidth requests are unknown to the SDN controller, one can use a suitable approach to learn them, and then use our BR policy using those learned parameters. Given these rates, there still remains the problem of determining the BR thresholds; Section V-B concerns an approximate approach for this problem. One could also propose online algorithms to learn the BR thresholds, e.g., see [54] where a stochastic approximation based algorithm was used to learn an optimal threshold-based policy for MDPs. In this paper, we do not address this question of learning the unknown parameters or the BR thresholds. This is an interesting future direction. *Dynamic routing:* Although we consider the fixed-routing model (which is relevant SDNs where routing decisions are made based on static link costs [55]), it would be interesting to study the case of dynamic routing, where bandwidth requests specify their source and destination, and the SDN controller gets to choose a route for the incoming bandwidth request

[8]Our policy `BR-EFP` does not require that the requests have equal mean holding times or equal bandwidth requirements.

[9]This need not be an optimal policy on link j when the requests have unequal mean holding times or unequal bandwidth requirements. In such cases, it is known that BR policies need not be optimal [18, page 1060].

if it gets accepted. Structural results for optimal admission control and routing in such settings is another interesting future direction.

B. `BR-EFP-APPROX`*: Approximate Reservation Thresholds for the k-hop Tandem Network Under Assumption (A0)*

The `BR-EFP` algorithm involves the computation of optimal BR thresholds on each link, the exact calculation of which is computationally hard. In this section, we formulate an optimization problem to identify an approximate BR threshold on each link in the k-hop problem (recall this from Section II-A) in another asymptotic regime where the arrival rates and link bandwidths become large. Throughout this section, we assume (A0): i.e., $\lambda_i = \lambda$, $r_i = 1$ for all $1 \le i \le k$, and that all the requests have unit mean holding times and unit bandwidth requirements. Let r denote the revenue rate of the k-hop request. While the methodology in this section can be adapted to more general networks, we focus on the above symmetric k-hop problem for ease of exposition. Since a BR policy is optimal on a single link [9], we reserve bandwidth (resp. do not reserve bandwidth) for either the λ-request[10] or the ν-request and we call it the high (resp. low) priority request.

Two scalings[11] of the network parameters are considered that result in different blocking probabilities as the scaling parameter becomes large: (i) the blocking probability of the high (resp. low) priority request vanishes (resp. lie strictly between 0 and 1); and (ii) the blocking probability of the high (resp. low) priority request vanishes (resp. becomes close to 1). In both these situations, we formulate an optimization problem to identify an approximate BR threshold for the low priority request.

1) Case I: $r < k$: First we study the case $r < k$. In this case, the revenue per hop for the k-hop requests is $\frac{r}{k} < 1$ (the revenue for the one-hop requests), and we reserve bandwidth for one-hop requests [9]. Consider a single isolated link in the k-hop tandem network, of bandwidth C. Given a BR threshold t, we consider the policy where we accept a λ-request (resp. ν-request) whenever the residual bandwidth is greater than or equal to 1 (resp. greater than or equal to t).[12] Under this policy, let B_λ (respectively B_ν) denote the blocking probability of a λ-request (respectively ν-request). By analyzing the birth and death process on the link, we have [56]

$$\begin{aligned}\frac{1}{B_\lambda} &= \frac{1}{E(\lambda, C)} - F(\lambda, C, t-1)\\ &\quad\left(\frac{1}{E(\lambda, C-t)} - \frac{\lambda}{\lambda+\nu}\frac{1}{E(\lambda+\nu, C-t)}\right);\\ \frac{1}{1-B_\nu} &= \frac{\lambda+\nu}{\lambda B_\lambda}\frac{E(\lambda+\nu, C-t)}{F(\lambda, C, t-1)},\end{aligned} \tag{13}$$

where $E(\lambda, C)$ is the Erlang blocking formula given by $E(\lambda, C) = \frac{\lambda^C/C!}{\sum_{i=0}^C \lambda^i/i!}$, and $F(\lambda, C, t-1) = \frac{C!}{\lambda^{t+1}(C-t)!}$. Since the direct calculation of the BR threshold t is hard, we consider the asymptotic regime $\lambda \to \infty$ and obtain the BR thresholds in that regime. Towards this, we carry out the calculations for the following special relationships between the arrival rates and the link bandwidths:

$$C = \lambda - \alpha\sqrt{\lambda},\ \ \alpha \in \mathbb{R},\ \text{and}\ \nu = \gamma\sqrt{\lambda},\ \ \gamma > 0. \tag{14}$$

The scaling of C in (14) is motivated by the "critical loading" regime in circuit switched networks (e.g., [56], [57] and the references therein). We let ν to grow like $O(\sqrt{\lambda})$ for concreteness. As we shall see, the scalings in (14) lead to B_λ going to 0 and B_ν going to a number strictly between 0 and 1 as $\lambda \to \infty$. One can also study other scalings, e.g., $\nu = O(\lambda)$; in that case, the approximations of the blocking probabilities in the sequel have to be modified appropriately.

Approximate blocking probabilities: Under the relationships given in (14), we can approximate the Erlang blocking formula $E(\lambda, C)$ as $\frac{1}{E(\lambda,C)} = \lambda^{1/2}W_0(\alpha)\left(1 + O\left(\frac{1}{\sqrt{\lambda}}\right)\right)$, where $W_0(\alpha) = \int_0^\infty e^{-(u^2/2)-\alpha u}du$ [56], [57]. Recall that our goal is to find an optimal BR threshold t. It is known that the optimal threshold under the scaling (14) grows as $O(\sqrt{\lambda})$ for large λ [58]. Therefore, let us write the BR threshold t in terms of another parameter β, i.e., we write $t = (\beta - \alpha)\sqrt{\lambda}$, so that the rejection threshold is $C - t = \lambda - \beta\sqrt{\lambda}$. By using the asymptotic approximation for the Erlang blocking formula, we get the following expression for another term in (14):

$$\frac{1}{E(\lambda, C-t)} = \lambda^{1/2}W_0(\beta)\left(1 + O\left(\frac{1}{\sqrt{\lambda}}\right)\right). \tag{15}$$

Consider the `BR-EFP` algorithm. Let $\hat{\nu}$ denote the thinned arrival rate of the k-hop request at the output of the algorithm, i.e., $\hat{\nu} = (1 - B_{\hat{\nu}})^{k-1}\nu$, where $B_{\hat{\nu}}$ is the blocking probability of the k-hop request on a link under the BR policy computed using the thinned arrival rate $\hat{\nu}$ (i.e., $B_{\hat{\nu}}$ is computed using (13) where ν is replaced by $\hat{\nu}$). Let $\hat{\gamma} = (1 - B_{\hat{\nu}})^{k-1}\gamma$, so that we have $\hat{\nu} = \hat{\gamma}\sqrt{\lambda}$.

Observe that $\frac{\lambda}{\lambda+\hat{\nu}} \sim 1$ for large λ, and $\lambda + \nu - (C - t) = (\beta + \hat{\gamma})\sqrt{\lambda}$. Thus, by using the asymptotic approximation of Erlang formula, we get $\frac{1}{E(\lambda+\hat{\nu}, C-t)} = \lambda^{1/2}W_0(\beta + \hat{\gamma})\left(1 + O\left(\frac{1}{\sqrt{\lambda}}\right)\right)$; $F(\lambda, C, t - 1) = e^{(\alpha^2-\beta^2)/2}\left(1 + O\left(\frac{1}{\sqrt{\lambda}}\right)\right)$ [56]. Using the above, (15), and $\frac{\lambda}{\lambda+\nu} \sim 1$, (13) becomes

$$\begin{aligned}B_\lambda &= \frac{\lambda^{-1/2}e^{-\alpha^2/2}}{\Delta(\alpha, \beta, \hat{\gamma})}\left(1 + O\left(\frac{1}{\sqrt{\lambda}}\right)\right)\ \text{and}\\ 1 - B_{\hat{\nu}} &= \frac{e^{-\beta^2/2}W_0(\beta + \hat{\gamma})}{\Delta(\alpha, \beta, \hat{\gamma})}\left(1 + O\left(\frac{1}{\sqrt{\lambda}}\right)\right),\end{aligned} \tag{16}$$

where $\Delta(\alpha, \beta, \hat{\gamma}) := e^{-\alpha^2/2}W_0(\alpha) - e^{-\beta^2/2}W_0(\beta) + e^{-\beta^2/2}W_0(\beta + \hat{\gamma})$. Note that $B_\lambda = O(1/\sqrt{\lambda})$ and $1 - B_{\hat{\nu}} = O(1)$, which is a consequence of the scaling of ν in (14).[13]

Identification of an optimal BR threshold: Having approximated B_λ and $B_{\hat{\nu}}$, we now obtain an approximate BR

[10] We refer to the one-hop request on an isolated link as the λ-request, and the k-hop request as the ν-request.

[11] We remark that the scalings considered in this section are specified using certain interrelationship among the arrival rates and the link bandwidths, unlike the scaling considered in (1).

[12] In the notation given in [56], a ν-request is accepted when the residual bandwidth is strictly greater than t.

[13] On the other hand, if we had $\nu = O(\lambda)$ instead of $O(\sqrt{\lambda})$, then a similar analysis shows that both B_λ and $1 - B_\nu$ (computed using (13)) are $O(1/\sqrt{\lambda})$.

threshold on each link. Using Little's theorem (e.g., [45, page 276]), the long-run average revenue on a link is given by $(1-B_\lambda)\lambda + (1-B_{\hat{\nu}})\hat{\nu}(r/k)$, where the thinned arrival rate $\hat{\nu}$ is given by the fixed point equation $\hat{\gamma} = (1-B_{\hat{\nu}})^{k-1}\gamma$. Recall that $t = (\beta-\alpha)\sqrt{\lambda}$. Thus, an approximate BR threshold t is obtained by maximizing the above long-run revenue expression w.r.t. β subject to the fixed point equation $\hat{\gamma} = (1-B_{\hat{\nu}})^{k-1}\gamma$. Thus, using (16), we have the following constrained optimization problem:

$$\min_{\beta\geq\alpha} \frac{e^{-\alpha^2/2} - e^{-\beta^2/2}W_0(\beta+\hat{\gamma})\hat{\gamma}(r/k)}{\Delta(\alpha,\beta,\hat{\gamma})}, \tag{17a}$$

$$\text{subject to } \hat{\gamma} = \left(\frac{e^{-\beta^2/2}W_0(\beta+\hat{\gamma})}{\Delta(\alpha,\beta,\hat{\gamma})}\right)^{k-1}\gamma; \quad \beta\geq\alpha. \tag{17b}$$

If β^* is a solution to the above problem, then an optimal BR threshold is $(\beta^*-\alpha)\sqrt{\lambda}$. The above problem is a non-linear optimization problem and the constraints are given by a fixed point equation in $\hat{\gamma}$. In Section VI, we solve the above problem numerically to obtain an approximation for an optimal bandwidth reservation threshold.

2) Case II: $r > k$: In this case, an optimal policy on each link is a BR policy where we reserve bandwidth for the k-hop request [9]. We consider the critical scaling (with $\hat{\nu}$ denoting the thinned arrival process of the k-hop requests):

$$C = \hat{\nu} - \alpha\sqrt{\hat{\nu}}, \;\; \alpha\in\mathbb{R}, \;\text{and}\;\; \lambda = \gamma\hat{\nu}, \;\; \gamma > 0. \tag{18}$$

We consider the scaling $\lambda = \gamma\hat{\nu}$ for concreteness. Similar to the $r < k$ case, it is straightforward to write down the approximate Erlang blocking formulas. We find that the blocking probability of the ν-request goes to 0 and that of the λ-request goes to 1 as $\nu\to\infty$. The revenue maximization problem in this case becomes:

$$\max_{\beta\geq\alpha} \;\hat{\nu}\frac{r}{k} + \frac{\hat{\nu}^{1/2}e^{-\beta^2/2}}{\Delta(\alpha,\beta)} - \frac{\hat{\nu}^{1/2}e^{-\alpha^2/2}(r/k)}{\Delta(\alpha,\beta)}$$

$$\text{subject to } \hat{\nu} = \left(1 - \frac{e^{-\alpha^2/2}}{\hat{\nu}^{1/2}\Delta(\alpha,\beta)}\right)^{k-1}\nu; \quad \beta\geq\alpha.$$

where the BR threshold is given by $t = (\beta-\alpha)\sqrt{\hat{\nu}}$, and $\Delta(\alpha,\beta) = e^{-\alpha^2/2}W_0(\alpha) - e^{-\beta^2/2}W_0(\beta)$.

VI. Numerical Results

In this section, we conduct several numerical experiments to evaluate the performance of the heuristic algorithms proposed in the previous section. The BR-EFP and BR-EFP-APPROX policies were introduced in Section V. Let BR-EFP-ES denote the BR-EFP algorithm in which, at each iteration, the optimal thresholds on each link are obtained by exhaustive search. Also, let OPT denote the optimal network-wide policy (which may not be a BR policy) for the MDP (in Section II) computed using the value iteration algorithm.

We study BR-EFP-APPROX, BR-EFP-ES, and OPT by conducting the following numerical experiments.

- We first evaluate the performance of BR-EFP-APPROX and compare it with BR-EFP-ES for k-hop tandem networks.
- Next, for small-sized (2-hop and 5-hop) tandem networks, we compare OPT and BR-EFP-ES. We find that the performance of BR-EFP-ES matches closely with that of OPT; this match turns out to be better for the 5-hop problems.
- Finally, we implement the BR-EFP-ES policy in general networks with more number of flows and larger link bandwidths. For such networks, it is intractable to compute OPT. We report the network-wide average revenue for the following: a k-hop tandem network, a star topology, a grid topology, and Google's B4 backbone topology [3].

TABLE III
BR-EFP-APPROX Vs. BR-EFP-ES for Two-Hop Tandem Networks With $r=1$, $\alpha=1$, and $\gamma=1$. $C-t$ Is the Rejection Threshold for the Two-Hop Requests

λ	ν	C	$C-t$ BR-EFP-ES	$C-t$ BR-EFP-APPROX	Revenue BR-EFP-ES	Revenue BR-EFP-APPROX
10	3.16	6	1	1	5.1549	5.1549
50	7.07	42	30	31	39.1245	39.1245
100	10	90	73	74	85.3903	85.3904
200	14.14	185	161	163	178.3977	178.3977
500	22.36	477	438	441	466.038	466.038
1000	31.62	968	913	918	952.1217	952.1217

TABLE IV
BR-EFP-APPROX Vs. BR-EFP-ES for Two-Hop Tandem Networks With $r=4$, $\alpha=1$, and $\gamma=1$. $C-t$ Is the Rejection Threshold for the One-Hop Requests

λ,ν	C	$C-t$ BR-EFP-ES	$C-t$ BR-EFP-APPROX	Revenue BR-EFP-ES	Revenue BR-EFP-APPROX
20	18	14	13	29.3939	29.3558
50	45	37	36	79.9885	79.8728
100	90	77	76	166.5211	166.5073
200	180	159	158	342.6217	342.6143
500	450	410	408	877.0895	877.0855
1000	900	834	829	1.7764×10^3	1.7728×10^3

A. BR-EFP-APPROX and BR-EFP-ES

We begin with the performance evaluation of the BR-EFP-APPROX policy studied in Section V-B for k-hop tandem networks. Recall the parameters $\alpha,\gamma > 0$, and the scaling considered in (14) and (18). We consider the symmetric k-hop problem where each one-hop request has arrival rate λ, and consider various problem instances by choosing the arrival rates λ,ν, and the link bandwidths C. For each problem instance, we compute two thresholds on a single link: (i) an optimal threshold given by BR-EFP-ES, and (ii) the approximate threshold given by BR-EFP-APPROX. For both these thresholds, we compare the corresponding average revenue from a single link.

We consider two cases. First, we consider the case $r < k$, the scaling in (14), and $\alpha=\gamma=1$. In this case, we reserve bandwidth for one-hop requests. Table III shows the rejection threshold $C-t$ (where t is the BR threshold according to Definition 1) and the average revenues obtained from BR-EFP-ES and BR-EFP-APPROX. Next, we consider the case $r > k$, the scaling (18), and $\alpha=\gamma=1$. Table IV summarizes the results for a two-hop network with $r=4$. Tables V and VI show numerical results for a five-hop network with $r=4$ and a four-hop network with $r=8$. We find that the bandwidth reservation thresholds as well as the revenues obtained from BR-EFP-APPROX are within 0.5% of that of BR-EFP-ES in all the experiments.

B. BR-EFP-ES and OPT

In the next set of experiments, we compare the performance of the BR-EFP-ES policy with OPT for 2-hop and 5-hop

TABLE V

BR-EFP-APPROX Vs. BR-EFP-ES FOR FIVE-HOP TANDEM NETWORKS WITH $r = 4$, $\alpha = 1$, AND $\gamma = 1$. $C - t$ IS THE REJECTION THRESHOLD FOR THE FIVE-HOP REQUESTS

λ	ν	$C-t$ BR-EFP-ES	$C-t$ BR-EFP-APPROX	Revenue BR-EFP-ES	Revenue BR-EFP-APPROX
10	3.16	5	6	5.1669	5.1549
50	7.07	39	41	39.1929	39.1585
100	10	86	89	85.5415	85.4438
200	14.14	179	183	178.8211	178.4741
500	22.36	466	473	466.367	466.199
1000	31.62	953	962	952.6357	952.369

TABLE VI

BR-EFP-APPROX Vs. BR-EFP-ES FOR FOUR-HOP TANDEM NETWORKS WITH $r = 8$, $\alpha = 1$, AND $\gamma = 1$. $C - t$ IS THE REJECTION THRESHOLD FOR THE ONE-HOP REQUESTS

λ, ν	C	$C-t$ BR-EFP-ES	$C-t$ BR-EFP-APPROX	Revenue BR-EFP-ES	Revenue BR-EFP-APPROX
500	450	433	434	826.4957	826.4696
1000	900	880	882	1.6365×10^3	1.6362×10^3

TABLE VII

COMPARISON OF BR-EFP-ES AND OPT FOR SEVERAL TWO-HOP PROBLEMS. ALL BANDWIDTH REQUESTS ARE ASSUMED TO BE FOR 1 UNIT, AND ALL MEAN HOLDING TIMES ARE ASSUMED TO BE 1 UNIT

Bandwidth of every link	Arrival rates $(\lambda_1, \lambda_2, \nu)$	Revenue rates (r_1, r_2, r_3)	Network revenue BR-EFP-ES	Network revenue OPT	Difference (in %)
5	1, 1, 3	1, 1, 1	3.14	3.91	19.69
5	3, 1, 1	1, 1, 4	6.10	6.61	7.72
5	1, 2, 3	2, 1, 1	4.4	5.27	16.51
5	3, 4, 1	1, 1, 5	7.42	8.72	14.91
5	1, 1, 1	1, 1, 2	2.61	2.87	9.06
5	1, 1, 1	1, 1, 2	3.23	3.8	15.0
10	1, 1, 1	1, 1, 1	2.86	2.99	4.35
10	1, 1, 1	1, 1, 4	5.69	5.99	5.01
10	1, 1, 3	1, 1, 1	3.68	4.96	25.81
10	3, 1, 1	1, 1, 4	6.88	7.96	13.57
10	1, 2, 3	2, 1 ,1	6.05	6.89	12.19
10	3, 4, 1	1, 1, 5	9.12	11.79	23.81

TABLE VIII

COMPARISON OF BR-EFP-ES AND OPT FOR SEVERAL FIVE-HOP PROBLEMS. ALL BANDWIDTH REQUESTS ARE ASSUMED TO BE FOR 1 UNIT, AND ALL MEAN HOLDING TIMES ARE ASSUMED TO BE 1 UNIT

Bandwidth of every link	Arrival rates $(\lambda_1, \ldots, \lambda_5, \nu)$	Revenue rates $(r_1, \ldots, r_5, r_6)$	Network revenue BR-EFP-ES	Network revenue OPT	Difference (in %)
2	1, 1, 1, 1, 1, 1	1, 1, 1, 1, 1, 6	5.14	5.56	7.55
2	1, 1, 1, 1, 1, 1	1, 1, 1, 1, 1, 3	4.37	4.38	0.23
2	3, 1, 2, 1, 1, 2	1, 2, 1, 4, 1, 2	8.21	8.21	0.0
2	1, 3, 1, 2, 1, 4	4, 1, 2, 1, 1, 3	8.21	8.24	0.36
4	1, 1, 1, 1, 1, 1	1, 1, 1, 1, 1, 6	8.64	9.24	6.49
4	1, 1, 1, 1, 1, 1	1, 1, 1, 1, 1, 3	6.22	6.88	9.59
4	3, 1, 2, 1, 1, 2	1, 2, 1, 4, 1, 2	11.43	11.8	3.14
4	1, 3, 1, 2, 1, 4	4, 1, 2, 1, 1, 3	12.3	13.3	7.52

tandem networks. In Table VII, we consider a class of two-hop problems and report the average long-run revenue obtained from BR-EFP-ES and OPT. In these experiments, we fix the bandwidths for all the links, assume a mean holding time of 1 unit and a bandwidth requirement of 1 unit for all the requests. The columns 'Arrival rates' (resp. 'Revenues') list the arrival rates (resp. revenue rates) of all the one-hop requests followed by the k-hop request. The average revenue obtained from BR-EFP-ES is comparable to that of OPT for a wide range of parameters: we find that, on an average, the network revenue from BR-EFP-ES is within 13% of OPT. Table VIII provides a similar comparison for a class of five-hop problems. We find that, on an average, the network revenue from BR-EFP-ES for the five-hop problems studied here is within 4.4% of OPT. This seems to suggests that the BR-EFP-ES policy gives a better network revenue for problems with a large state space.

C. BR-EFP-ES for General Networks

Finally, we implement BR-EFP-ES for the following network topologies: the 2-hop and 5-hop tandem networks in Figure 1, a star topology consisting of 7 nodes and bandwidth requests on 6 routes, a grid topology consisting of 16 nodes and bandwidth requests on 6 routes, and Google's B4 backbone topology consisting of 12 nodes and bandwidth requests on 5 routes (see Figure 4). We assume that each link in every network has a bandwidth of 10 units, each request has a bandwidth requirement of 1 unit, and a mean holding time of 1 unit. For each topology, we first implement BR-EFP-ES and obtain the network-level policy. We then obtain the average revenue from this policy by simulating the network for 2 million discrete events and computing the average revenue over time. The outcomes of this numerical study are summarized in Table IX. We observe that, for many examples, the network revenue from BR-EFP-ES is very close to the LP upper bound in (3).

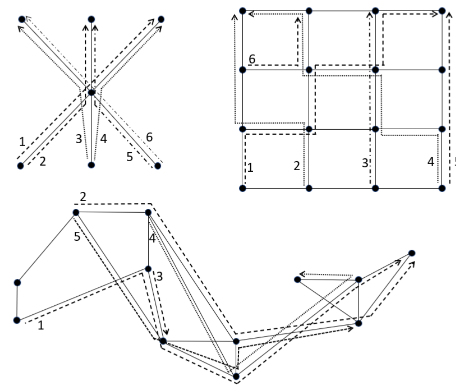

Fig. 4. The network topologies considered for the performance evaluation of BR-EFP-ES. Bandwidth requests are numbered starting from 1, and the dashed/dotted lines indicate their routes.

TABLE IX

BR-EFP-ES FOR GENERAL NETWORKS. FOR THE k-HOP PROBLEMS, THE ARRIVAL RATES AND REVENUE RATES ARE LISTED AS $(\lambda_1, \ldots, \lambda_k, \nu)$ AND $(r_1, \ldots, r_k, r_{k+1})$, RESPECTIVELY. FOR THE OTHER TOPOLOGIES, THE ARRIVAL RATES AND REVENUE RATES ARE LISTED IN THE SAME ORDER AS SHOWN IN FIGURE 4

Network topology	Arrival rates on the various routes	Revenue rates	Network revenue BR-EFP-ES	LP upper bound in (3)	Difference (in %)
2-hop	1,4,1	5,4,2	22.61	23	1.7
2-hop	1,4,1	5,1,2	10.84	11	1.45
5-hop	1,2,1,3,1,1	1,2,2,1,1,25	35.95	36	0.14
5-hop	1,1,1,1,1	1,2,3,1,5,7	19.0	19	0.0
Star	1,1,2,2,1,3	10,12,2,3,2,2	39.68	40	0.80
Star	1,1,2,1,4,3	1,3,2,1,2,4	27.17	29	6.31
Grid	1,2,1,3,1,2	8,12,2,3,2,20	84.08	85	1.08
Grid	1,2,1,3,1,2	8,2,3,2,10,1	32.72	33	0.85
B4	4,1,2,1,1	15,4,20,2,8	107.56	114	5.65
B4	1,1,1,1,1	20,6,2,5,10	42.91	43	0.21

VII. CONCLUSION

This paper has reported a study of bandwidth reservation policies in the context of the Bandwidth on Demand service in Software Defined Networks. Revenue optimal admission control can be formulated as a Markov decision problem; such formulations suffer from the curse of dimensionality, and, typically, do not lead to explicit solutions. We first identified an optimal policy for a k-hop tandem network when each link has unit bandwidth. We then considered the k-hop tandem network in the "Kelly" limiting regime where the arrival rates and the link bandwidths become large. One of the main theoretical results of this paper is that, under certain parameter regimes, a policy of BR form is asymptotically optimal for the k-hop tandem network. The main ingredient in the proof of this result is the unique identification of certain acceptance probabilities of requests in the limiting regime. Motivated by this result, we then proposed a BR-type policy (BR-EFP-ES) for the BoD problem in general networks and evaluated its performance numerically on both k-hop tandem networks and more general multihop networks. Finally, we studied an optimization problem to approximately compute the BR thresholds for the k-hop tandem network (BR-EFP-APPROX) and evaluated its performance numerically.

The following could be fruitful directions for further research on the BoD problem. Bandwidth reservation is a well-known technique for preventing instability in alternate routing of virtual circuits; the problem of revenue maximization with multiple classes of bandwidth requests in such networks could be explored. For developing a learning approach to bandwidth reservation a possible direction could be that of Roy et al. [54], where reinforcement learning is used to learn a threshold policy for a Markov decision problem. It would also be interesting to study the BoD problem in a non-asymptotic regime (i.e., for a fixed network) and to compare the performance of the best BR policy with the optimal revenue.

Appendix A
Proofs of Section III

A. Preliminary Lemmas

Lemma 1: Let $\pi_A^{(1)} : \mathcal{S} \times \{1,2\} \to \{0,1\}$ be a policy specified by a set $A \subseteq \{0,1,\cdots,k\}$. Then, either A is of the form $A = \{p, p+1, \dots, k\}$ for some $0 \le p \le k$ or there exists a policy π' specified by a set A' of the above form such that $\mathcal{R}(\pi) = \mathcal{R}(\pi')$.

Proof: Under any admissible policy π, note that the process $\{X(t), t \ge 0\}$ is a finite-state Markov chain with non-zero and finite transition rates and hence the empty state is regenerative. Also, observe that, under any admission control policy, the process $X(t)$ has no two disjoint closed sets, and hence the long-run average revenue under policy π is independent of the initial state (Theorem 7.1.1 of [59]). Thus, we can fix the initial state to be the empty state.

If A is of the given form, there is nothing to prove. Else, define

$$p' := \min\{i \mid 0 \le i \le k,\, i \in A\} \tag{19}$$

and consider the policy π' specified by the set $\{p', p'+1, \dots, k\}$. Starting from the empty state, observe that the evolution of the process $X(t)$ under both π and π' are the same (sample path-wise), and hence it follows that the average revenue under π and π' are the same. □

Lemma 2: We have $\mathcal{R}(\pi^{(0)}) = \frac{k\lambda}{\lambda+1}$, $\mathcal{R}(\pi_0^{(1)}) = \frac{r\nu}{\nu+1}$, and $\mathcal{R}(\pi_l^{(1)}) = \frac{\nu r + \sum_{i=1}^{l} i\binom{k}{i}\lambda^i}{\nu+1+\sum_{i=1}^{l}\binom{k}{i}\lambda^i}$, $1 \le l \le k$.

Proof: Under $\pi^{(0)}$, since we do not accept a k-hop request, each link behaves independently. By applying the renewal reward theorem on each link and summing up over all links, we get $\mathcal{R}(\pi^{(0)}) = k \times \frac{1}{\frac{1}{\lambda}+1} = \frac{k\lambda}{\lambda+1}$. Under $\pi_0^{(1)}$, since we do not accept a one-hop request, the renewal reward theorem gives $\mathcal{R}(\pi_0^{(1)}) = \frac{r}{\frac{1}{\nu}+1} = \frac{\nu r}{\nu+1}$.

Fix $l \in \{1,2,\dots,k\}$. Let ζ denote the equilibrium distribution of the process $X(t)$ under policy $\pi_l^{(1)}$. Then, it is straightforward to compute from the detailed balance equations that

$$\zeta(0,0) = \frac{1}{1+\nu+\binom{k}{1}\lambda+\cdots+\binom{k}{l}\lambda^l},$$

$$\zeta(i,0) = \begin{cases} \frac{\binom{k}{i}\lambda^i}{1+\nu+\binom{k}{1}\lambda+\cdots+\binom{k}{l}\lambda^l}, & 1 \le i \le l, \\ 0, & l+1 \le i \le k, \end{cases}$$

$$\zeta(0,1) = \frac{\nu}{1+\nu+\binom{k}{1}\lambda+\cdots+\binom{k}{l}\lambda^l}.$$

In state $(i,0)$, $0 \le i \le k$, we get a revenue rate of i, and in state $(0,1)$ we get a revenue rate of r. Thus, the average revenue under policy $\pi_k^{(1)}$ is

$$\mathcal{R}(\pi_l^{(1)}) = \sum_{i=0}^{l} i\zeta(i,0) + r\zeta(0,1) = \frac{\nu r + \sum_{i=1}^{l} i\binom{k}{i}\lambda^i}{\nu+1+\sum_{i=1}^{l}\binom{k}{i}\lambda^i}.$$

□

Lemma 3: Suppose that $\mathcal{R}(\pi_l^{(1)}) > \mathcal{R}(\pi_{l-1}^{(1)})$ for some $1 \le l < k$. Then $\mathcal{R}(\pi_{l+1}^{(1)}) \ge \mathcal{R}(\pi_l^{(1)})$.

Proof: Write $\mathcal{R}(\pi_l^{(1)})$ as $\mathcal{R}(\pi_l^{(1)}) = \frac{\nu r + x_l}{\nu + y_l}$, where $x_l = \sum_{i=1}^{l} i\binom{k}{i}\lambda^i$ and $y_l = 1 + \sum_{i=1}^{l}\binom{k}{i}\lambda^i$. Suppose that $\mathcal{R}(\pi_l^{(1)}) \ge \mathcal{R}(\pi_{l-1}^{(1)})$ for some $1 \le l < k$, i.e.,

$$\frac{\nu r + x_l}{\nu + y_l} \ge \frac{\nu r + x_{l-1}}{\nu + y_{l-1}},$$

which after rearrangements give

$$\nu(x_l - x_{l-1}) + \nu r(y_{l-1} - y_l) + x_l y_{l-1} - x_{l-1} y_l \ge 0. \tag{20}$$

Notice that we can write $x_l = x_{l-1} + p_{l-1}$, where $p_{l-1} = l\binom{k}{l}\lambda^l$. Similarly, $y_l = y_{l-1} + q_{l-1}$, where $q_{l-1} = \binom{k}{l}\lambda^l$. Also, we have $p_l = \frac{\lambda(k-l)}{l} p_{l-1}$, and $q_l = \frac{\lambda(k-l)}{l+1} q_{l-1}$. Therefore, (20) becomes

$$\nu(p_{l-1} - r q_{l-1}) + p_{l-1} y_{l-1} - q_{l-1} x_{l-1} \ge 0. \tag{21}$$

Recall that our goal is to show $\mathcal{R}(\pi_{l+1}^{(1)}) \ge \mathcal{R}(\pi_l^{(1)})$, i.e., to show $\nu(p_l - r q_l) + p_l y_l - q_l x_l \ge 0$. Consider the first term. By writing p_l and q_l in terms of p_{l-1} and q_{l-1} respectively, we have $\nu(p_l - r q_l) = \nu\lambda(k-l)(\frac{p_{l-1}}{l} - \frac{r q_{l-1}}{l+1}) \ge \frac{\nu\lambda(k-l)}{l}(p_{l-1} - r q_{l-1})$. Also, note that $p_l y_l - q_l x_l = \lambda(k-l)\left(\frac{p_{l-1}}{l}(y_{l-1} + q_{l-1}) - \frac{q_{l-1}}{l+1}(x_{l-1} + p_{l-1})\right) \ge \frac{\lambda(k-l)}{l}(p_{l-1} y_{l-1} - q_{l-1} x_{l-1})$. Thus, $\nu(p_l - r q_l) + p_l y_l - q_l x_l \ge \frac{\lambda(k-l)}{l}\left(\nu(p_{l-1} - r q_{l-1}) + p_{l-1} y_{l-1} - q_{l-1} x_{l-1}\right) \ge 0$, where the inequality follows from (21). □

B. Proof of Theorem 1

We compare the average revenues $\mathcal{R}(\pi^{(0)})$, $\mathcal{R}(\pi_k^{(1)})$ and $\mathcal{R}(\pi_0^{(1)})$ to obtain the structure of optimal policy.

$\pi^{(0)}$ is optimal when $\mathcal{R}(\pi^{(0)}) \ge \mathcal{R}(\pi_k^{(1)})$ and $\mathcal{R}(\pi^{(0)}) \ge \mathcal{R}(\pi_0^{(1)})$. Note that $\mathcal{R}(\pi^{(0)}) \ge \mathcal{R}(\pi_k^{(1)})$ implies that $\frac{k\lambda}{\lambda+1} \ge \frac{\nu r + k\lambda(\lambda+1)^{k-1}}{\nu + (\lambda+1)^k}$, i.e., $r \le \frac{k\lambda}{\lambda+1}$. Also, $\mathcal{R}(\pi^{(0)}) \ge \mathcal{R}(\pi_0^{(1)})$ implies that $\frac{k\lambda}{\lambda+1} \ge \frac{\nu r}{\nu+1}$, i.e., $r \le \frac{k\lambda}{\lambda+1}\left(\frac{\nu+1}{\nu}\right)$. Combining the above, we see that $\pi^{(0)}$ is optimal if $r \le \frac{k\lambda}{\lambda+1}$.

Similarly, $\pi_k^{(1)}$ is optimal when $\mathcal{R}(\pi_k^{(1)}) \ge \mathcal{R}(\pi^{(0)})$ and $\mathcal{R}(\pi_k^{(1)}) \ge \mathcal{R}(\pi_0^{(1)})$. It is straightforward to check that $\mathcal{R}(\pi_k^{(1)}) \ge \mathcal{R}(\pi^{(0)})$ implies $r \ge \frac{k\lambda}{\lambda+1}$, and $\mathcal{R}(\pi_k^{(1)}) \ge \mathcal{R}(\pi_0^{(1)})$

implies $r \leq \frac{k\lambda}{\lambda+1}\left(\frac{\nu+1}{\nu}\right)\left(\frac{(1+\lambda)^k}{(1+\lambda)^k-1}\right)$. Thus, $\pi_k^{(1)}$ is optimal if $\frac{k\lambda}{\lambda+1} \leq r \leq \frac{k\lambda}{\lambda+1}\left(\frac{\nu+1}{\nu}\right)\left(\frac{(1+\lambda)^k}{(1+\lambda)^k-1}\right)$.

Similarly, $\pi_0^{(1)}$ is optimal when $\mathcal{R}(\pi_0^{(1)}) \geq \mathcal{R}(\pi^{(0)})$ and $\mathcal{R}(\pi_0^{(1)}) \geq \mathcal{R}(\pi_k^{(1)})$. Once again, it is straightforward to check that $\mathcal{R}(\pi_0^{(1)}) \geq \mathcal{R}(\pi^{(0)})$ implies $r \geq \frac{k\lambda}{\lambda+1}\left(\frac{\nu+1}{\nu}\right)$, and $\mathcal{R}(\pi_0^{(1)}) \geq \mathcal{R}(\pi_k^{(1)})$ implies $r \geq \frac{k\lambda}{\lambda+1}\left(\frac{\nu+1}{\nu}\right)\left(\frac{(1+\lambda)^k}{(1+\lambda)^k-1}\right)$. Combining the above two conditions, we see that $\pi_0^{(1)}$ is optimal if $r \geq \frac{k\lambda}{\lambda+1}\left(\frac{\nu+1}{\nu}\right)\left(\frac{(1+\lambda)^k}{(1+\lambda)^k-1}\right)$. This completes the proof of the theorem.

Appendix B
Proofs of Section IV

A. Proof of Theorem 2

By Theorem 3 in [17] and its extension to the case when the bandwidth reservation parameters vary with N (see Section III in [17]), we get that $\{\mathbf{x}^{(N)}\}$ is relatively compact in $D_{\mathcal{X}}[0,\infty)$ and any subsequential weak limit of $\{\mathbf{x}^{(N)}\}$ must satisfy

$$x_j(t) = x_j(0) + \int_0^t (\lambda_j P_j(\mathbf{x}(u)) - x_j(u))du,$$

$j \in \{1,2,\ldots,k+1\}$ and $t \geq 0$, where $P_j(\mathbf{x})$, $\mathbf{x} \in \mathcal{X}$ and $j \in \{1,2,\ldots,k+1\}$, satisfy the following conditions:

(C1) $P_{k+1}(\mathbf{x}) = \eta_1^{\mathbf{x}}(\{\mathbf{m} : m_i \geq 1 \text{ for all } i\})$, where $\eta_1^{\mathbf{x}}$ is some stationary distribution of the process $\mathbf{m}_1^{\mathbf{x}}$ defined in (11) such that, for each $i \in \{1,2,\ldots,k\}$, $\eta_1^{\mathbf{x}}(\{\mathbf{m} : m_i = \infty\}) = 1$ if $x_i + x_{k+1} < C_i$.

(C2) For each $j \in \{1,2,\ldots,k\}$, $P_j(\mathbf{x}) = \eta_2^{\mathbf{x}}(\{\mathbf{m} : m_j \geq 0\})$, where $\eta_2^{\mathbf{x}}$ is some stationary distribution of the process $\mathbf{m}_2^{\mathbf{x}}$ defined in (12) such that, for each $i \in \{1,2,\ldots,k\}$, $\eta_2^{\mathbf{x}}(\{\mathbf{m} : m_i = \infty\}) = 1$ if $x_i + x_{k+1} < C_i$.

Note that the above result in [17] asserts the convergence of $\{\mathbf{x}^{(N)}\}$ along subsequences. In the rest of the proof, we show that $P_j(\mathbf{x})$, $j \in \{1,2,\ldots,k+1\}$ and $\mathbf{x} \in \mathcal{X}$, are given as in the statement of the theorem. Once the acceptance probabilities P_j and P_{k+1} are uniquely identified regardless of the subsequence, then the result follows from standard results in weak convergence theory (see, e.g., [49, Chapter 3]).

To show $P_{k+1}(\mathbf{x}) = 1$ for all $\mathbf{x} \in \mathcal{X}$: If $\mathbf{x}$ is such that $x_j + x_{k+1} < C_j$ for all $j \in \{1,2,\ldots,k\}$, then noting that the distribution $\eta_1^{\mathbf{x}}$ of $\mathbf{m}_1^{\mathbf{x}}$ defined by $\eta_1^{\mathbf{x}}(\{+\infty,\ldots,+\infty\}) = 1$ satisfies condition (C1), and it follows that $P_{k+1}(\mathbf{x}) = 1$. Thus we focus on an $\mathbf{x}$ such that, for some nonempty $A \subset \{1,2,\ldots,k\}$, $x_j + x_{k+1} = C_j$ for all $j \in A$. Consider the Markov chain $\mathbf{m}_1^{\mathbf{x}}$ defined in (11). We show that any stationary distribution $\eta_1^{\mathbf{x}}$ of the Markov chain (11) such that $\eta_1^{\mathbf{x}}(\{\mathbf{m} : m_i = \infty\}) = 1$ for all $i \notin A$ (which is required by (C1)) is the points mass at $(+\infty,+\infty,\ldots,+\infty)$. Once this is shown, condition (C1) will then imply that $P_{k+1}(\mathbf{x}) = 1$.

Towards this, for $A' \subset A$, consider the Markov chain $\mathbf{m}_1^{A'}$ with state space $\mathbb{Z}_+^{|A'|}$ obtained from the chain $\mathbf{m}_1^{\mathbf{x}}$ in (11) by freezing all its components outside A' to $+\infty$ (we suppress the dependence of $\mathbf{m}_1^{A'}$ on $\mathbf{x}$ for ease of readability). This chain $\mathbf{m}_1^{A'}$ has the following transition rates:

$$\mathbf{m}_1^{A'} \rightarrow \begin{cases} \mathbf{m}_1^{A'} + \mathbf{1}_{A'} & \text{at rate } x_{k+1}, \\ \mathbf{m}_1^{A'} + \mathbf{e}_j & \text{at rate } x_j,\ j \in A', \\ \mathbf{m}_1^{A'} - \mathbf{1}_{A'} & \text{at rate } \nu, \end{cases}$$

where $\mathbf{1}_{A'}$ denotes the vector of length $|A'|$ with a 1 in each position corresponding to A'. Define the function f on $\mathbb{Z}_+^{|A'|}$ by $f(\mathbf{m}) = \sum_{i\in A'} m_i$, $\mathbf{m} \in \mathbb{Z}_+^{|A'|}$. Note that its drift at a state $\mathbf{m}$ with $m_i > 0$ for all $i \in A'$ is

$$\begin{aligned}\sum_{m'}&(f(\mathbf{m}') - f(\mathbf{m})) \times \text{rate}(\mathbf{m} \rightarrow \mathbf{m}') \\ &= |A'|x_{k+1} + \sum_{i\in A'} x_i \\ &\quad - |A'|\nu = \sum_{i\in A'}(x_i + x_{k+1} - \nu).\end{aligned}$$

Since $x_i + x_{k+1} = C_i$ for all $i \in A'$, by Proposition 1, we find that $(x_i + x_{k+1} - \nu) \geq \lambda_i \geq 0$ for all $i \in A'$. On the other hand, given a state $\mathbf{m}$ such that $m_i = 0$ for some $i \in A'$, we find that the drift of the function f at $\mathbf{m}$ is bounded below by the quantity in the RHS of the previous display. In all cases, the drift is nonnegative and uniformly bounded above. Hence, the embedded discrete-time Markov chain corresponding to $\mathbf{m}_1^{A'}$ is either null recurrent or transient (see, e.g., [45, Chapter I, Proposition 5.4]). Since the transition rates are uniformly bounded, the Markov chain $\mathbf{m}_1^{A'}$ is either null recurrent or transient (see, e.g., [45, Section II.4]), and hence it does not possess any stationary distribution. That is, any stationary distribution $\eta_1^{\mathbf{x}}$ of the process $\mathbf{m}_1^{\mathbf{x}}$ such that $\eta_1^{\mathbf{x}}(\{\mathbf{m} : m_i = \infty\}) = 1$ for all $i \notin A$ is supported on $(+\infty,+\infty,\ldots,+\infty)$. Therefore, by condition (C1), we conclude that $P_{k+1}(\mathbf{x}) = 1$.

To show $P_j(\mathbf{x})$ in (10): Fix $j \in \{1,2,\ldots,k\}$ and consider $P_j(\mathbf{x})$. If $\mathbf{x}$ is such that $x_j + x_{k+1} < C_j$, then defining the stationary distribution $\eta_2^{\mathbf{x}}$ of $\mathbf{m}_2^{\mathbf{x}}$ by $\eta_2^{\mathbf{x}}(\{+\infty,\ldots,+\infty\}) = 1$, it follows from condition (C2) that $P_j(\mathbf{x}) = 1$. We now consider the case when $x_j + x_{k+1} = C_j$. In this case we identify $P_j(\mathbf{x})$ by observing that the limiting dynamical system must always belong to $\mathcal{X}$. Towards this, fix $\mathbf{x} \in \mathcal{X}$ such that $x_j + x_{k+1} = C_j$ and consider the Markov chain $\mathbf{m}_2^{\mathbf{x}}$ on $(\mathbb{Z} \cup \{+\infty,-\infty\})^k$ defined by (12). We first define a stationary distribution $\eta_2^{\mathbf{x}}$ of this chain using the j-th component of $\mathbf{m}_2^{\mathbf{x}}$. Define the one-dimensional Markov chain m_2^j on $\mathbb{Z} \cup \{+\infty,-\infty\}$ (which is the Markov chain $\mathbf{m}_2^{\mathbf{x}}$ restricted to the j-th link with all the other components frozen at either $+\infty$ or $-\infty$) with the following transition rates:

$$m_2^j \rightarrow \begin{cases} m_2^j + 1 & \text{at rate } x_{k+1}, \\ m_2^j + 1 & \text{at rate } x_j, \\ m_2^j - 1 & \text{at rate } \nu, \\ m_2^j - 1 & \text{at rate } \lambda_j \times 1_{\{m_2^j \geq 0\}}. \end{cases}$$

Define the distribution η_2^j on $\mathbb{Z} \cup \{+\infty,-\infty\}$ as follows: (i) if $\nu = C_j$ (which corresponds to $\bar{x}_j = 0$), then define $\eta_2^j(\{-\infty\}) = 1$; (ii) if $\nu < C_j$, there are two possible situations: (a) if $\lambda_j + \nu > C_j$, then define η_2^j as the unique stationary distribution of the chain m_2^j restricted to $\mathbb{Z}$, (b) if $\lambda_j + \nu \leq C$ then define $\eta_2^j(\{+\infty\}) = 1$. In both cases, η_2^j is a stationary distribution of the Markov chain m_2^j on

$\mathbb{Z} \cup \{+\infty, -\infty\}$. Clearly, in (i), we have $\eta_2^j(\{m : m \geq 0\}) = 0$. In (ii)a, using the detailed balance equation, it can be checked that $\eta_2^j(\{m : m \geq 0\}) = \frac{x_j + x_{k+1} - \nu}{\lambda_j} = \frac{C_j - \nu}{\lambda_j}$, and in (ii)b, we have $\eta_2^j(\{m : m \geq 0\}) = 1$. Therefore, in all cases, we have $\eta_2^j(\{m : m \geq 0\}) = \frac{C_j - \nu}{\lambda_j} \wedge 1$.

We now define a stationary distribution $\eta_2^{\mathbf{x}}$ of $\mathbf{m}_2^{\mathbf{x}}$. If $\mathbf{m} \in (\mathbb{Z} \cup \{+\infty, -\infty\})^k$ is such that

- $m_i = +\infty$ for all $i \neq j$, define $\eta_2^{\mathbf{x}}(\mathbf{m}) = \eta_2^j(m_j)$;
- $m_i < \infty$ for some $i \neq j$, define $\eta_2^{\mathbf{x}}(\mathbf{m}) = 0$.

By condition (C2), $\eta_2^{\mathbf{x}}(\{\mathbf{m} : m_j \geq 0\})$ is a candidate for $P_j(\mathbf{x})$. By the definition of $\eta_2^{\mathbf{x}}$, this quantity is the same as $\eta_2^j(\{m : m \geq 0\})$, and it is given by (10). We now argue that this is the only possible choice of $P_j(\mathbf{x})$. Let $\tilde{\eta}_2^{\mathbf{x}}$ be another stationary distribution of $\mathbf{m}_2^{\mathbf{x}}$ such that $P_j(\mathbf{x}) = \tilde{\eta}_2^{\mathbf{x}}(\{\mathbf{m} : m_j \geq 0\})$. On one hand, we cannot have $\tilde{\eta}_2^{\mathbf{x}}(\{\mathbf{m} : m_j \geq 0\}) > \frac{C_j - \nu}{\lambda_j}$ since the value of $x_j(t) + x_{k+1}(t)$ of any limiting dynamics $\mathbf{x}(\cdot)$ that satisfies (9) must be at most C_j at all times t. On the other hand, we cannot have $\tilde{\eta}_2^{\mathbf{x}}(\{\mathbf{m} : m_j \geq 0\}) < \frac{C_j - \nu}{\lambda_j}$ since, by condition (C2), a trajectory starting at an $\mathbf{x}' \in \mathcal{X}$ such that $x'_j + x'_{k+1} = C_j$ cannot satisfy (9). Hence we conclude that $P_j(\mathbf{x})$ is given by (10).

We have thus identified the acceptance probabilities $P_j(\mathbf{x})$, $j \in \{1, 2, \ldots, k+1\}$ and $\mathbf{x} \in \mathcal{X}$, uniquely, and this completes the proof of the theorem.

B. Proof of Proposition 2

Let $\mathbf{x}(\cdot)$ be a trajectory that satisfies (9) in Theorem 2. It can be written in differential form as follows:

$$\dot{x}_{k+1}(t) = \nu - x_{k+1}(t),\ t \geq 0,$$
$$\dot{x}_j(t) = \begin{cases} \lambda_j - x_j(t) & \text{if } x_j(t) + x_{k+1}(t) < C_j, \\ (C_j - \nu) \wedge \lambda_j - x_j(t) & \text{if } x_j(t) + x_{k+1}(t) = C_j, \end{cases}$$

$j \in \{1, 2, \ldots, k\}$ and $t \geq 0$. Let $\mathbf{x}^*$ be a limit point of this trajectory; a limit point exists since $\mathcal{X}$ is compact. We will show that

$$\mathbf{x}^*(j) = \begin{cases} \nu, & \text{if } j = k+1, \\ \lambda_j, & \text{if } \bar{x}_j = \lambda_j,\ j \in \{1, 2, \ldots, k\}, \\ C_j - \nu, & \text{if } 0 < \bar{x}_j < \lambda_j,\ j \in \{1, 2, \ldots, k\}, \\ 0, & \text{if } \bar{x}_j = 0,\ j \in \{1, 2, \ldots, k\}; \end{cases}$$

then the result follows from Proposition 1.

Clearly, from the evolution of $x_{k+1}(t)$, it follows that $x_{k+1}(t) \to \nu$ as $t \to \infty$. Consider $x_j(t)$, $j \in \{1, 2, \ldots, k\}$. If j is such that $\bar{x}_j = \lambda_j$, then $C_j - \nu \geq \lambda_j$ (see Proposition 1). Therefore, $\dot{x}_j(t) = \lambda_j - x_j(t)$ for all $t \geq 0$. We therefore conclude that $x_j(t) \to \lambda_j$ as $t \to \infty$. On the other hand, if j is such that $\bar{x}_j < \lambda_j$, we have $C_j - \nu < \lambda_j$ (from Proposition 1). Therefore, we have

$$\dot{x}_j(t) + \dot{x}_{k+1}(t) = \begin{cases} \lambda_j + \nu - (x_j(t) + x_{k+1}(t)) > 0, \\ \qquad\qquad \text{if } x_j(t) + x_{k+1}(t) < C_j, \\ 0, \qquad\quad\ \text{if } x_j(t) + x_{k+1}(t) = C_j, \end{cases}$$

for all $t \geq 0$. Hence, we have $x_j(t) + x_{k+1}(t) \to C_j$ as $t \to \infty$. From the already established fact that $x_{k+1}(t) \to \nu$ as $t \to \infty$, it follows that $x_j(t) \to C_j - \nu$ as $t \to \infty$. Finally, note that $C_j - \nu = 0$ if $\bar{x}_j = 0$. This completes the proof.

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